

ARCHAEOLOGICAL INVENTORY SURVEY
TMK: 2-1-3-02:002, EAST HONOMAELE,
HANA DISTRICT
ISLAND OF MAUI

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ISLAND OF MAUI

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SUMMARY

At the request of Mr. Rich Komen, Haun & Associates conducted an archaeological inventory survey of TMK: 2-1-3-02:002, a c. 125-acre parcel located in the Land of East Honomaele, Hana District, Island of Maui. The objective of the survey was to satisfy historic preservation regulatory review inventory requirements of the Department of Land and Natural Resources-Historic Preservation Division (DLNR-HPD), as contained within Hawaii Administrative Rules, Title 13, DLNR, Subtitle 13, State Historic Preservation Rules.

The survey identified four sites with seven features. The sites consist of two complexes of historic sugar cane plantation railroad features (Sites 4963 and 4964), a historic road (Site 4965), and a human skeleton (Site 4966). The skeletal remains represent an isolated late prehistoric to early historic burial. Site 4964, a railroad bed and bridge abutment, were constructed before 1915 and abandoned by the 1920s. A second railroad (Site 4963) consists of three stream crossings where bridge abutments and support piers are present. The second railroad was originally constructed between 1915 and 1923. There is evidence that the bridge support structures were subsequently repaired and rebuilt. The Site 4965 roadbed was probably constructed after the 1920s, possibly as late as the 1960s.

Sites 4963 and 4965 are assessed as significant under Criterion "d". The mapping, written description, and photography at the sites adequately document them and no further work or preservation is recommended. Site 4964 is assessed as significant under Criterion "d" and Criterion "c". The Feature B abutment is recommended for preservation. Site 4966 is assessed as significant under Criterion "d" and Criterion "e". The site is recommended for preservation.

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INTRODUCTION

This report presents the results of an archaeological inventory survey conducted of TMK: 2-1-3-02:002, located in the Land of East Honomaele, Hana District, Island of Maui (*Figure 1*). The objective of the survey was to satisfy current historic preservation regulatory review inventory requirements of the Department of Land and Natural Resources-Historic Preservation Division (DLNR-HPD), as contained within Hawaii Administrative Rules, Title 13, DLNR, Subtitle 13, State Historic Preservation Rules (DLNR 1998).

The survey fieldwork was conducted May 17-19, 2000. Described in this final report are the project scope of work, field methods, background information, survey findings, and significance assessments of the sites with recommended further treatments.

Scope Of Work

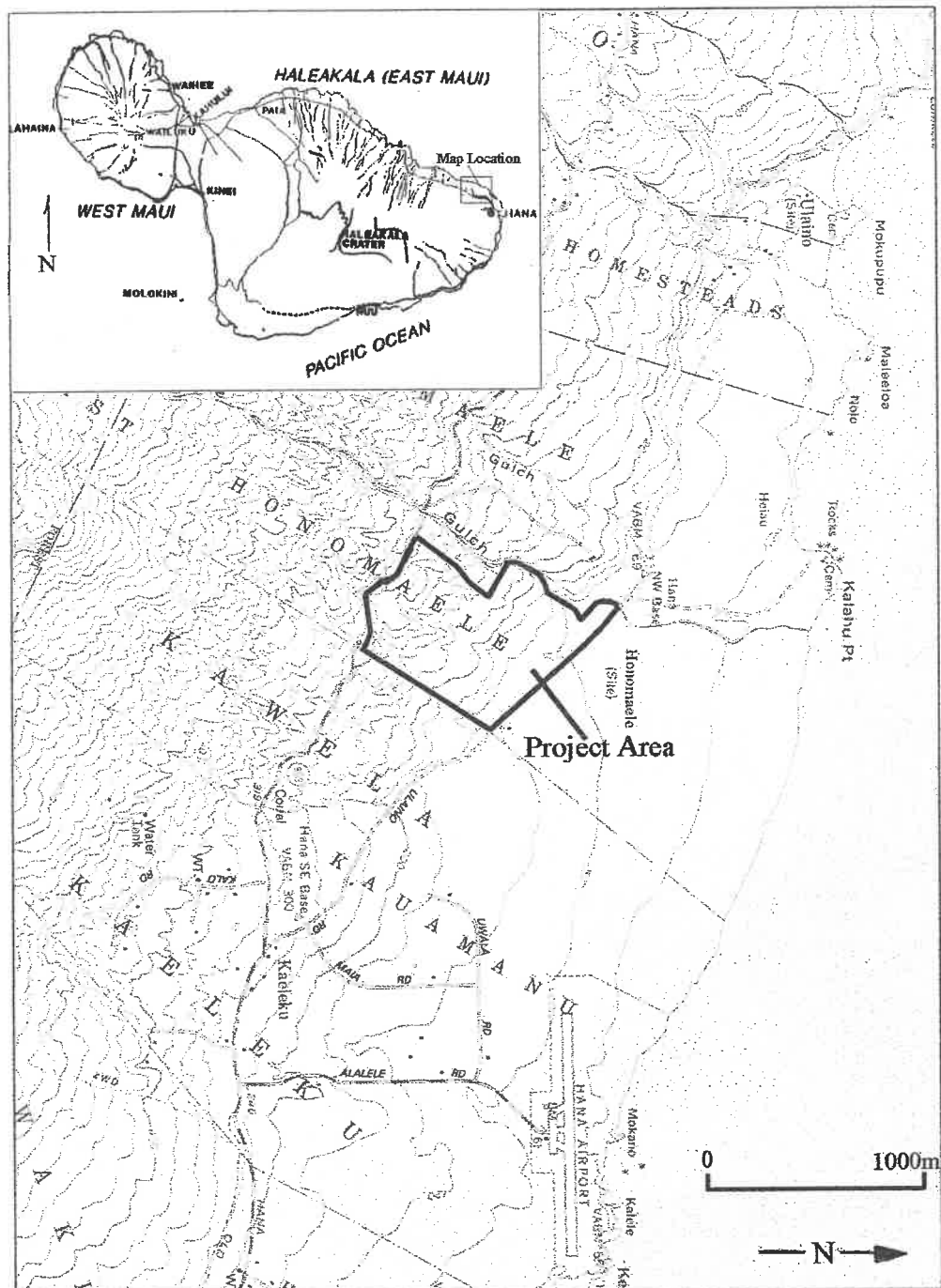
Based on DLNR-HPD rules for inventory surveys, the following specific tasks were determined to constitute an appropriate scope of work for the project:

1. Conduct background review and research of existing archaeological and historical documentary literature relating to the project area and its immediate vicinity--including examination of Land Commission Awards, *ahupua'a* records, historic maps, archival materials, archaeological reports, and other historical sources;
2. Conduct a high intensity, 100% pedestrian survey coverage of the project area;
3. Conduct detailed recording of all potentially significant sites including scale plan drawings, written descriptions, and photographs, as appropriate;
4. Conduct limited subsurface testing (manual excavation) at selected sites (a) to determine the presence or absence of potentially significant buried cultural deposits or features, and (b) to obtain suitable samples for radiocarbon age determination analyses;
5. Analyze background research and field data; and
6. Prepare and submit Final Report.

Project Area Description

The project area consists of a c. 125-acre parcel bounded on the north by Ulaino Road, on the south by the Hana Highway, on the east by a hog wire fence, and on the west by Honomaele Gulch and two lots (*Figure 1*). There are three main stream drainages that bisect the project area in a *mauka-makai* direction. These include Honomaele Gulch along the western project area boundary, an unnamed drainage in the approximate center of the parcel, and an unnamed drainage near the eastern boundary. The two unnamed drainages each have smaller branch drainages. The *makai* end of the easternmost drainage makes a sweeping bend to the northwest adjacent to Ulaino Road where the drainage meets the broad coastal plain. This portion of the drainage roughly parallels the northern project area boundary. The drainage eventually joins Honomaele Gulch, which also makes a sharp turn to the west at its junction with the coastal plain. The central *mauka-makai* drainage empties into the northern segment of the eastern drainage.

All of the drainages were dry during the survey. According to the landowner, Honomaele Gulch and the eastern drainage flow during wet weather. The central drainage flows during periods of heavy rainfall. The northern side of the project area between the base of the mountain slope and Ulaino Road is a



relatively level flood plain. The area is flooded by several feet of water following prolonged periods of heavy rain.

The project area ranges in elevation from 80 to 480 ft. The terrain slopes moderately to steeply toward the ocean. Rainfall in the general vicinity of the project area averages 75 inches per year and the mean annual temperature is 70 to 75 degrees F (Armstrong 1983). There are three soil types in the project area. Most of the area, approximately 95%, is comprised of Hana very stony silt clay loam (3-25 % slopes) (Foote et al. 1972). This soil type is characterized by a layer of dark brown to very dark grayish brown silt clay loam, over a subsoil comprised of a blocky dark brown silt clay loam. A substrate of weathered, pebble-sized cinders above a lava is present below the clay loam soils. This soil type evidences slow to medium runoff, with a slight to moderate erosion hazard.

The two remaining soil types consist of a second soil variant of the Hana series, and Rock land. A small area of Hana extremely stony silt clay loam on 3 to 25 % slopes is located in the northeastern portion of the project area. According to Foote et al., "this soil has a profile like that of Hana very stony silt clay loam, 3 to 25 % slopes, except that stones cover 3 to 15 percent of the surface" (1972:37). The Rock land series is located within Honomaele Gulch and is characterized by exposed basalt rock that covers 25 to 90% of the surface (Foote et al. 1972:119).

The project area has been extensively disturbed by historic and modern agricultural and ranching activity (Figures 2 and 3). Cattle were grazing in the parcel during the survey. The majority of the area evidences low grasses, with scattered stands of secondary vegetation primarily within and adjacent to the drainages. This vegetation consists of dense vines and ferns and a variety of species including African Tulip (*Spathodea campanulata*), bamboo (*Bambusa* sp), banana (*Musa spp.*), banyan (*Ficus benghalensis* L.), breadfruit (*Artocarpus communis*), coconut (*Cocos nucifera*), ginger (*Alpinia purpurata* Viell.; K. Schum.), guava (*Psidium cattleianum* f. *lucidum* Degener), hibiscus (*Hibiscus tiliaceus* L.), Java plum (*Eugenia cumini* [L.] Druce), candlenut (*Aleurites moluccana*), mango (*Manifera indica* L.), pandanus (*Pandanus odoratissimus* L.f.), papaya (*Carica papaya* L.), ti (*Cordyline terminalis* L. Kunth) and taro (*kalo*; *Colocasia esculenta* L. Schott). Macadamia nut (*Macadamia integrifolia* Maiden & Betche) orchards are present in the northwest and southwest corners of the project area.

Field Methods

As stated, the project area has been extensively impacted by historic and modern agricultural and ranching activity. The majority of the survey parcel consists of low grasses that have been cleared of vegetation by recent bulldozing and cattle grazing. As the ground surface visibility in these areas was excellent, these areas were examined through a series of *mauka-makai* transects with surveyors spaced at 30-50 m intervals. The three drainages in the project area were most densely vegetated and were examined with surveyors spaced at 5-10 m intervals.

The identified sites and features were flagged with blue and pink flagging tape and their locations plotted on a scaled project area map. The sites were recorded in detail, which consisted of the preparation of scaled plan maps, the completion of standardized site/feature forms, and photographic documentation.

Subsurface testing was undertaken in the lowland area near the northwestern boundary of the project area. The testing consisted of excavating five judgmentally placed shovel tests. The test units were excavated in stratigraphic layers and the removed soils screened through 1/8" mesh. Portable remains collected were placed in paper bags labeled with the appropriate provenience information.

An intact human skeleton was encountered during the excavation of one of the shovel tests. Upon identification of the human remains, the unit was expanded and its excavation continued with a trowel. Once articulation was determined, the remains were covered with a cotton cloth and the unit was carefully backfilled. Section drawings depicting the stratigraphy were prepared and the units were backfilled. Recovered cultural remains were transported to Haun & Associates laboratory for analysis.

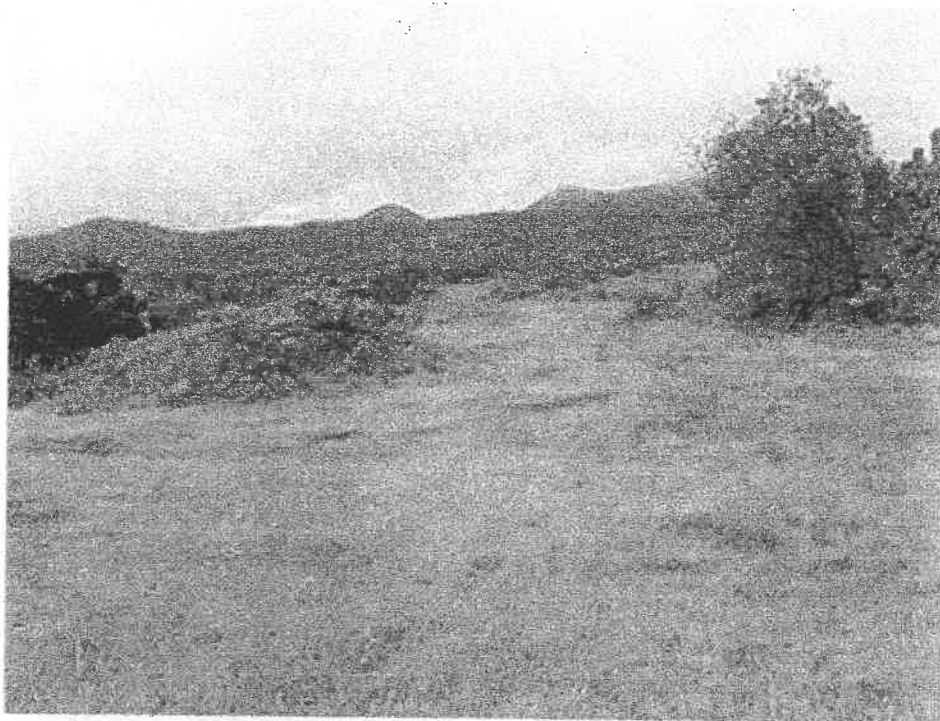


Figure 2. Project Area, view to south



Figure 3. Project Area, view to west

ARCHAEOLOGICAL AND HISTORICAL BACKGROUND

Historical Documentary Research

Archival research was conducted at the Hamilton Library Hawaii and Pacific Collection at the University of Hawaii-Manoa, the University of Hawaii-Hilo Hawaiian Collection, the Land Survey Office and the Archives Division of the Hawaii Department of Accounting and General Services, and the Hawaii State Public Libraries in Honolulu and Hilo.

Legendary accounts and traditional historical information concerning Hana District are described in detail by Cleghorn and Rogers (1987), Orr (in Kolb 1993), and Sterling (1998). Legends concerning the deities Pele, Pu'uhele, Kane, Kanaloa, Maui, and Ku'ula figure prominently in Hana's legendary history. Aiai, son of fish the god, Ku'ula, put three pebbles in the sea beyond the breaking waves off Honomaele. The pebbles attracted other pebbles and grew into a ridge. The underwater ridge became a fishing ground (*ko'a*) for *aweoweo*, providing a bountiful supply of fish for the people (Sterling (1998).

Honomaele and Kaeleku Ahupua'a were the source of 'ohia wood and ie'ie vines used to construct the ladders that provided access to the important chiefly residence and fortress at Ka'uiki Hill overlooking Hana Bay (Kamakau 1961). Kawaihau in Honomaele was the western edge of the battlefield for the Battle of Kapalipilo between the chief of Maui, Kamehameha-nui, and the chiefs of Hawaii. Piilanihale Heiau, on the coast of West Honomaele, is thought to be the residence of the Piilani line of Maui chiefs (Sterling 1998).

Captain Cook's ship *Resolution* stood off Hana's shore for four days in November 1778 (Barrow 1993). They "saw people on several parts of the shore, and some houses and plantations. The country seemed to be well wooded and watered." (1993:404). The Hawaiians traded cuttlefish, breadfruit, potatoes, taro, bananas and small pigs for nails and iron tools.

Handy and Handy (1972) report the Hana area was used to cultivate taro, yams, bananas, *wauke*, and *olona*. The area was famous for its 'awa. Handy and Handy relate stories about the chiefly settlement at Hana Bay. The area was favored because of its fishponds, fortress, and bountiful gardens. They report a coastal settlement at Hamoa in the 1930s where people raised sweet potatoes and obtained fish from the sea and a fishpond. Taro and bananas were cultivated in inland gardens. They also report a sizable settlement at Honokalani situated above the sea cliffs and fresh water caves of Waianapanapa. A small valley below Pu'u Olopawa at 1,500 ft elevation was previously used to grow taro in the dry season. A *hala* forest covered the coastal plain formed by recent lava flows between Ulaino and Hana. In 1934, dryland taro was still cultivated for subsistence in Honomaele.

The Waihona 'Aina database (Waihona 'Aina Corp. 1998) lists three Land Commission Award (LCA) claims within Honomaele. Two claims were awarded. Kahanu apparently claimed the entire *ahupua'a* under LCA Nos. 7774 and 42 M.A. Only 42 M.A. was awarded, which consisted of the *ahupua'a* of West Honomaele. The award documentation refers to it as "1/2 Honomaele Hana", Apana 1, consisting of 990 acres.

The other LCA is No. 4667 awarded to Kuahiwinui in the Ili of Kukuilona. The two-acre parcel is not shown on current tax maps. The claim testimony describes it as situated "*mauka* of the seashore and seaward of the Government Road and it lies as far as Pohaku Palahalaha in Waialele". The claim was disputed by William L. Davis. Kuahiwinui received the land before 1819. He had a house on the land, but did not live in it after 1841 or 1842. Afterwards, the house was occupied by a man employed by Kuahiwinui. Taro was cultivated on the Hana side of the parcel. Sugar cane was planted in several patches as early as 1846. The testimony also mentions the cultivation of beans.

No claims were made in the surrounding *ahupua'a* of Makapuu to the west and Kawela and Kaeleku to the east. The heirs of John Young claimed Ulaino situated on the west side of Makapuu; however, the claim was not awarded. LCA testimony from *ahupua'a* from Hana town to the southeast along the coast refer to the cultivation of sugar cane, wet and dry taro, sweet potatoes, and coconuts (Cleghorn and Rogers 1987). Pig husbandry is also mentioned.

Documents obtained or examined at the State Archives, including Privy Council minutes, a newspaper article, and Ministry of Interior correspondence (Appendix B), provide insight into land tenure. An article in the Advertiser of September 6, 1910 describes the Kaeleku Sugar Company as one of the oldest in the state. The article indicates that the predecessor of the company was started in 1852 by Cooke, Needham, and Unna.

Minutes from a Privy Council meeting on May 31, 1851 describe Resolution 11 in which the Minister of the Interior granted fee simple title to 441 acres of land in East Honomaele to W.G. Needham and E.J. Kelk. The grant reserved the rights of native tenants.

A letter dated April 25, 1876 to D. Uaiwa from Minister of Interior W.L. Moehonua refers to a letter from Uaiwa requesting a royal patent on 706 acres of land at Honomaele awarded to Kahanu under award 42. The letter requests documentation of the boundaries before granting the patent request.

A Department of Interior document dated July 21, 1893 by G.L. Brown describes government lease land in Honomaele, Kawela and Kaeleku. The lease area of 1,250 acres encompasses Kawela, East Honomaele, and most of Kaeleku between the present forest reserve boundary and the shoreline. The lease excluded various grants within the boundaries. *Figure 4* is a map of the lease area. The map shows Kelk and Needham as owners of land in East Honomaele, which includes the project area. The map also indicates that the original Government Road, named Okaka Puu Road, corresponds to the route of today's Ulaino Road. The upper road, which later became the Hana Highway, was not present in 1893.

A letter dated June 3, 1893 from M.H. Reuter to Minister of Interior J.A. King refers to the recent return of lease land in Kawela, Honomaele, and Kaeleku to the government. Reuter offers to pay an annual lease of \$100 for fifteen years. A letter dated January 8, 1894 from August Unna to Minister of Interior E.O. Hall transmits \$50 to pay lease rent on government lease land in Kawela, Honomaele, and Kaeleku for the two years ending August 26, 1872. The letter states Unna did not release the land in 1873 because he purchased 600 acres in Kaeleku. Unna offers to pay an annual lease of \$25, which was the amount paid previously, for ten years. The lease area extends from the government road to the ocean. A letter dated January 8, 1894 from Unna to Minister of Interior L.G. Wilder transmits \$25 to pay lease rent for the year of 1880.

A 1900 map of Honomaele (*Figure 5*) surveyed by Hugh Hoconnel(?) shows an upper road labeled "New Wagon Road to Nahiku". The lower road is labeled "wagon road" on the east side of Kawaikau Gulch (today's Honomaele Gulch), which divides East and West Honomaele. West of the gulch, the road is labeled "Horse Trail". The map indicates the area from just inland of the upper government road to the coast in most of East Honomaele is Grant 788.

A map surveyed by W.E. Wall and traced by H.E. Newton in 1915 (*Figure 6*) shows Kelk and Needham as owners of land in East Honomaele. A plantation railroad extends into the east edge of the project area. The map also shows the boundaries of General Lease 1227 to Kaeleku Sugar Company. The lease area indicated by a dashed line encompasses much of East Honomaele, Kawela and Kaeleku. The annual lease rent was \$1,380. The map also shows a small group buildings, two railroad spurs, and a school lot near the intersection of the upper and lower government roads in Kaeleku.

McCall (1940) describes the early development of commercial sugar cane cultivation in Hana. The Hana Plantation was managed by George W. Wilfong in 1851 with 60 cultivated acres. In 1852, he brought laborers from China. The small mill only produced syrup, which was sold to whale ships. Wilfong left the plantation after the mill burned and he could not obtain credit to rebuild.

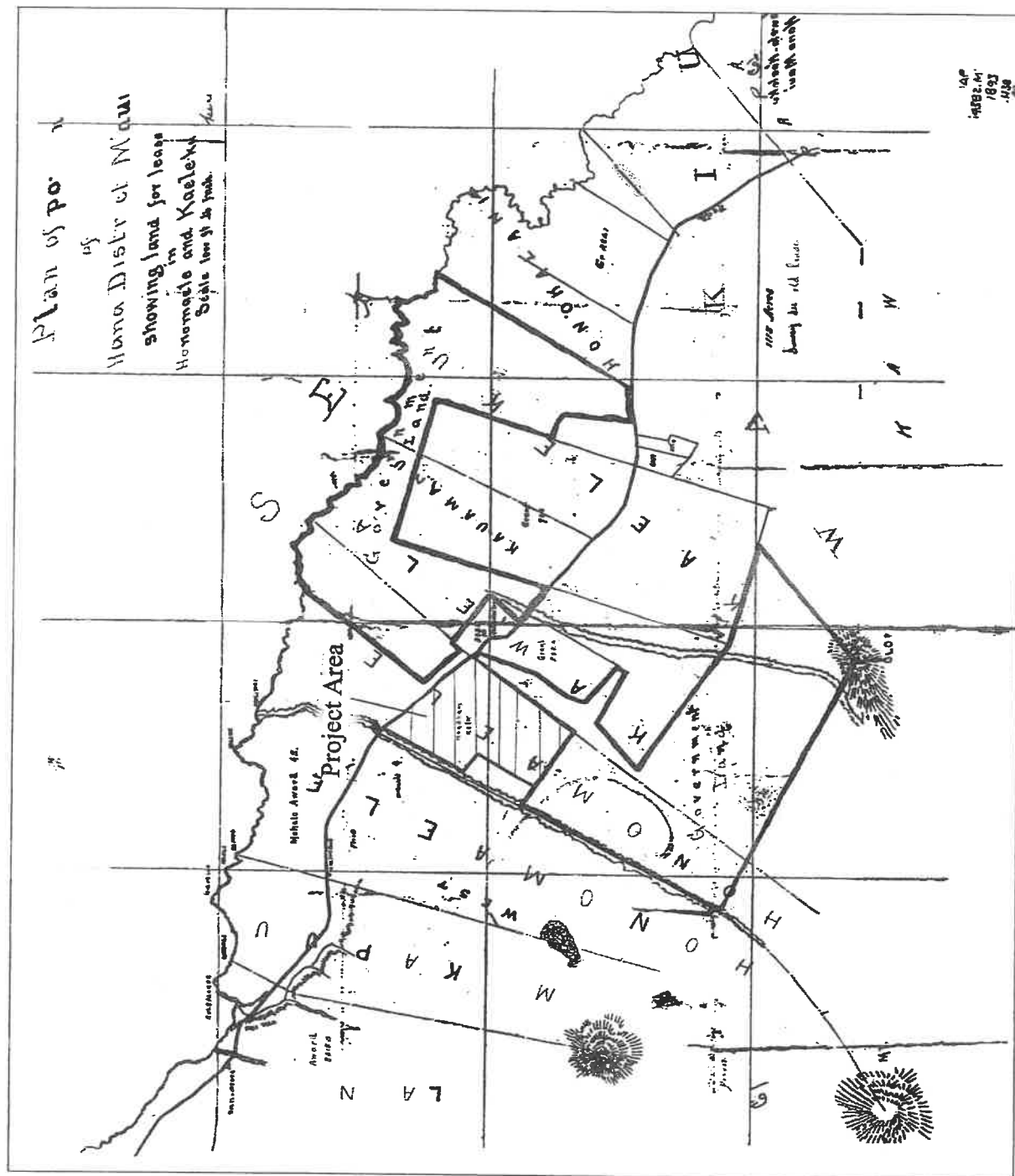


Figure 4. 1893 Map of Government Lease Lands

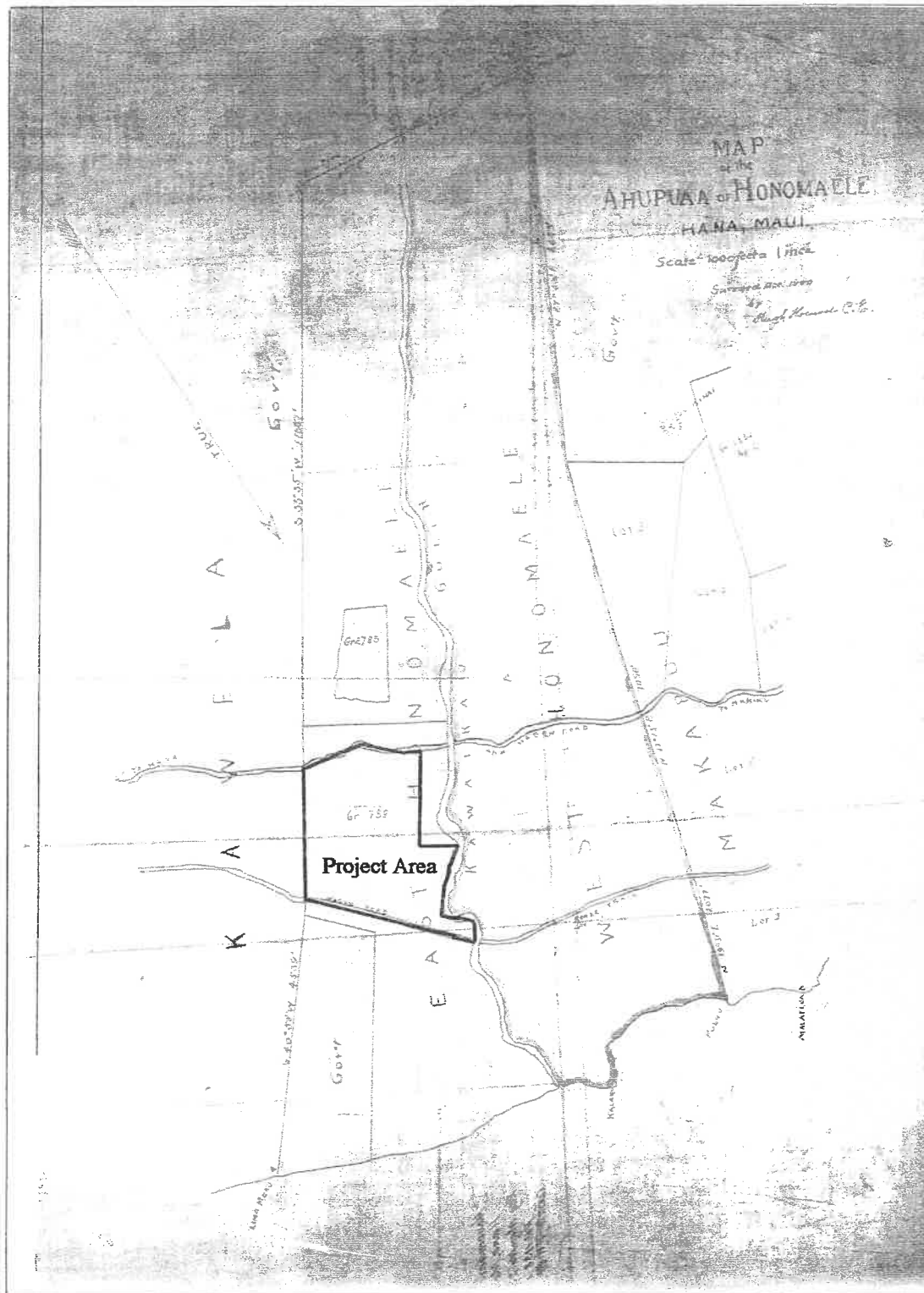
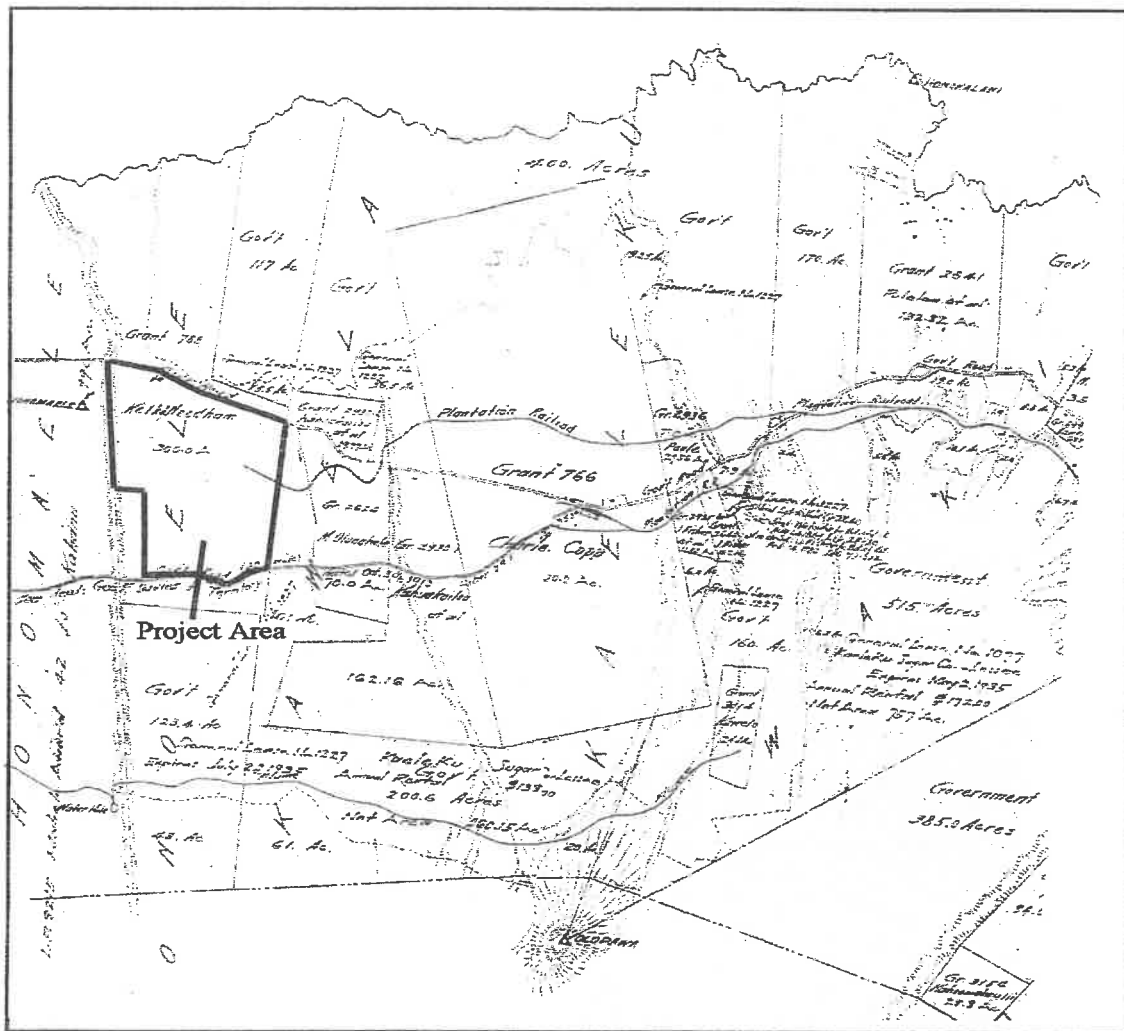


Figure 5. 1900 Honomaele Map



The lands of Hana Plantation passed through a succession of owners until the partnership of Thomas E. Cook, William G Needham, August Unna, a native of Denmark, controlled them in 1861 (Conde and Best 1973). Needham left the partnership soon after it was formed and Cook left in 1867. In 1868, Unna imported workers from Japan. Unna is credited with the development of the railroad system that was put in service in 1883. Unna died in 1885. Flumes were developed to provide water to the mills and to transport cane from the fields.

In 1883, the Reciprocity Sugar Company was founded and by 1888 the company owned 2,800 acres with 600 acres in cultivation and 240 employees (McCall 1940). In 1888, the Hana Sugar Company consisted of 5,000 acres with 700 in cultivation. The company had 250 employees and 250 head of working stock. M.S. Grinbaum formed the Hana Plantation Company in 1889, combining the lands of the Hana and Reciprocity Sugar Companies with lands at Hamoa (Conde and Best 1973).

The Kaeleku Sugar Company was established in 1905 (McCall 1940). The company took over the Hana Plantation lands, which consisted of 886 acres in fee and 13,184 in leasehold. In 1913, 300 acres were leased from the Hamoa Agricultural Company. Additional acreage was leased from the Haneoo Agricultural Company bringing the total acreage to 15,407 acres. Only about 20% of the land could be cultivated because of gulches and rocky areas. The Kaeleku Sugar Company eventually included the lands of six former plantations (McCall 1940). Only two plantations, Hana and Reciprocity, had mills and piers.

A US Coastal and Geodetic Survey (USCGS) map based on surveys between 1923 and 1925 (*Figure 7*) shows a plantation railroad track extending along the seaward edge of the project area. The track ends in West Honomalee at a small group of buildings. A second railroad spur terminates in a small group of structures in Kawela. This spur follows the alignment of the railroad tracks on the 1915 map (see *Figure 6*), but terminates before it reaches the project area. The USCGS map also shows a flume extending through the western side of the project area terminating at the railroad tracks.

In 1945, Kaeleku Plantation closed because of a lack of labor and the threat of unionization (Cleg-horn and Rogers 1987). In the same year, most of the property controlled by the plantation was purchased by one of the owners, Paul Fagan. Fagan converted the plantation to a cattle ranch and opened the Ka'uiki Inn, which later became Hotel Hana-Maui. Mechanized pasture improvement using bulldozers began in the 1960s.

Much of the land surrounding the project area still belongs to Hana Ranch. In the late 1980s, entertainer Jim Nabors purchased over 400 acres including the project area. The project area was subsequently sold to the present owner, Rich Komen. Harry Koon, a resident of Hana for the past 20 years who works for the present owner, recalls Nabors using a bulldozer to clear the project area. Koon reported that an elderly Hawaiian man called "Blue" told him graves were present in two locations in the project area. Other individuals told Koon that these areas had been bulldozed.

Previous Archaeological Research

A search of DLNR-HPD archaeological report database and other sources identified 25 reports for Hana District between Ulaino and Hamoa. *Figure 8* shows the locations of 15 survey projects and *Table 1* summarizes the projects. Not included in the figure and table are the general studies by Thrum (1917), Walker (1931), Nakkim (1970), Ashdown (1971), and Orr (1990), which focus on major sites, primarily *heiau* and fishponds, throughout Hana District, and a walk through survey by Sterling (1969) in Hana Town. Other site-specific studies not included are inspections of a lava tube system, Kaeleku Caverns, by Estioko-Griffin (1988) and Donham (1996). Burials are reported for the cave, but were not identified by the inspections. Kam (1980) conducted an inspection of areas surrounding Hana Airport with negative results. Dixon (1998) reported the discovery of an apparently isolated human cranium from a cinder quarry at the base of Pu'u Olopawa.

Orr (1990) reviewed previous studies by Walker (1931), Ashdown (1971), Nakkim (1970), and others in her report on *heiau* of Hana. She lists 34 *heiau* in the eleven miles of Hana's shoreline between Keaa Beach to

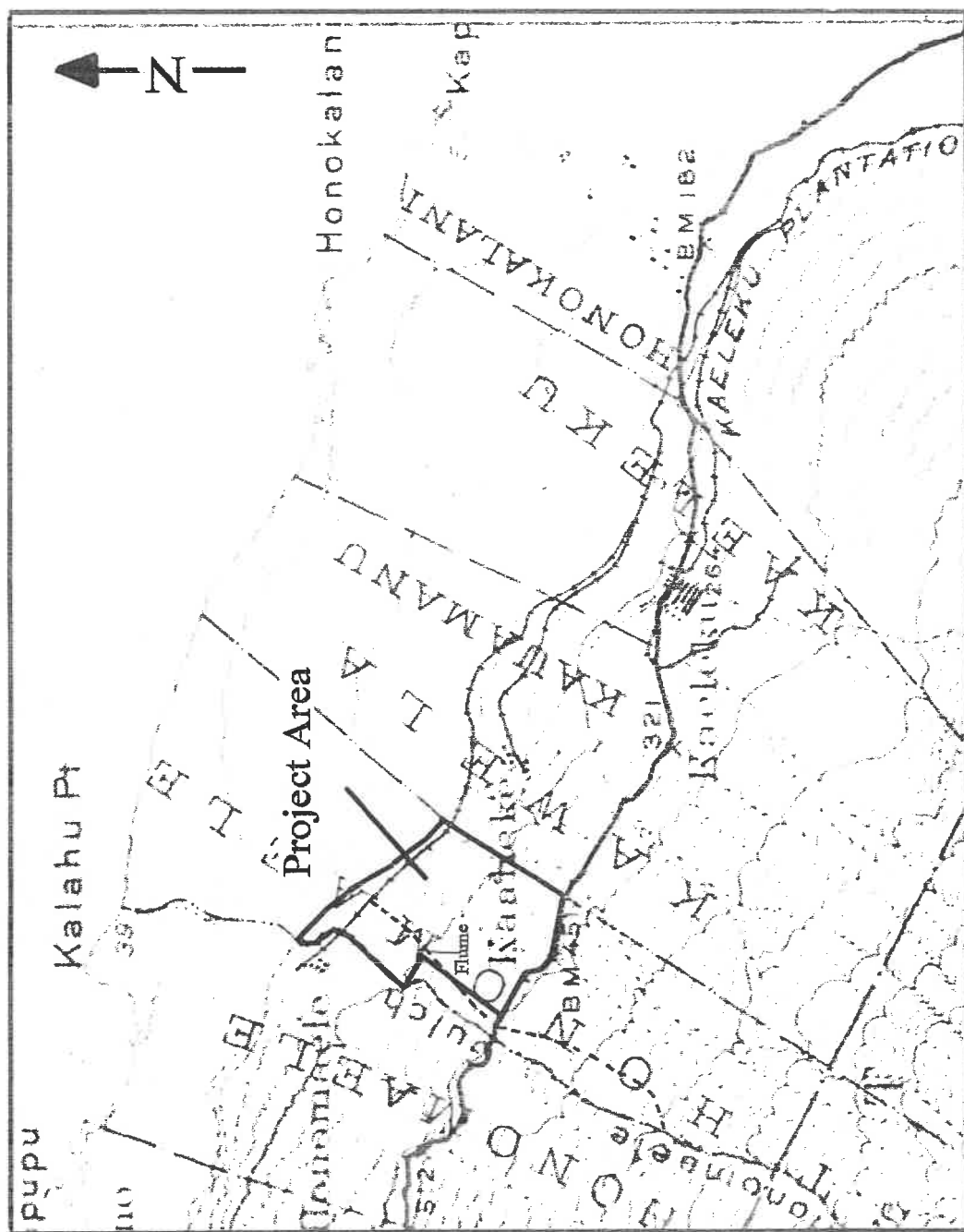


Figure 7. USCGS Hana Quadrangle (1923-1925)

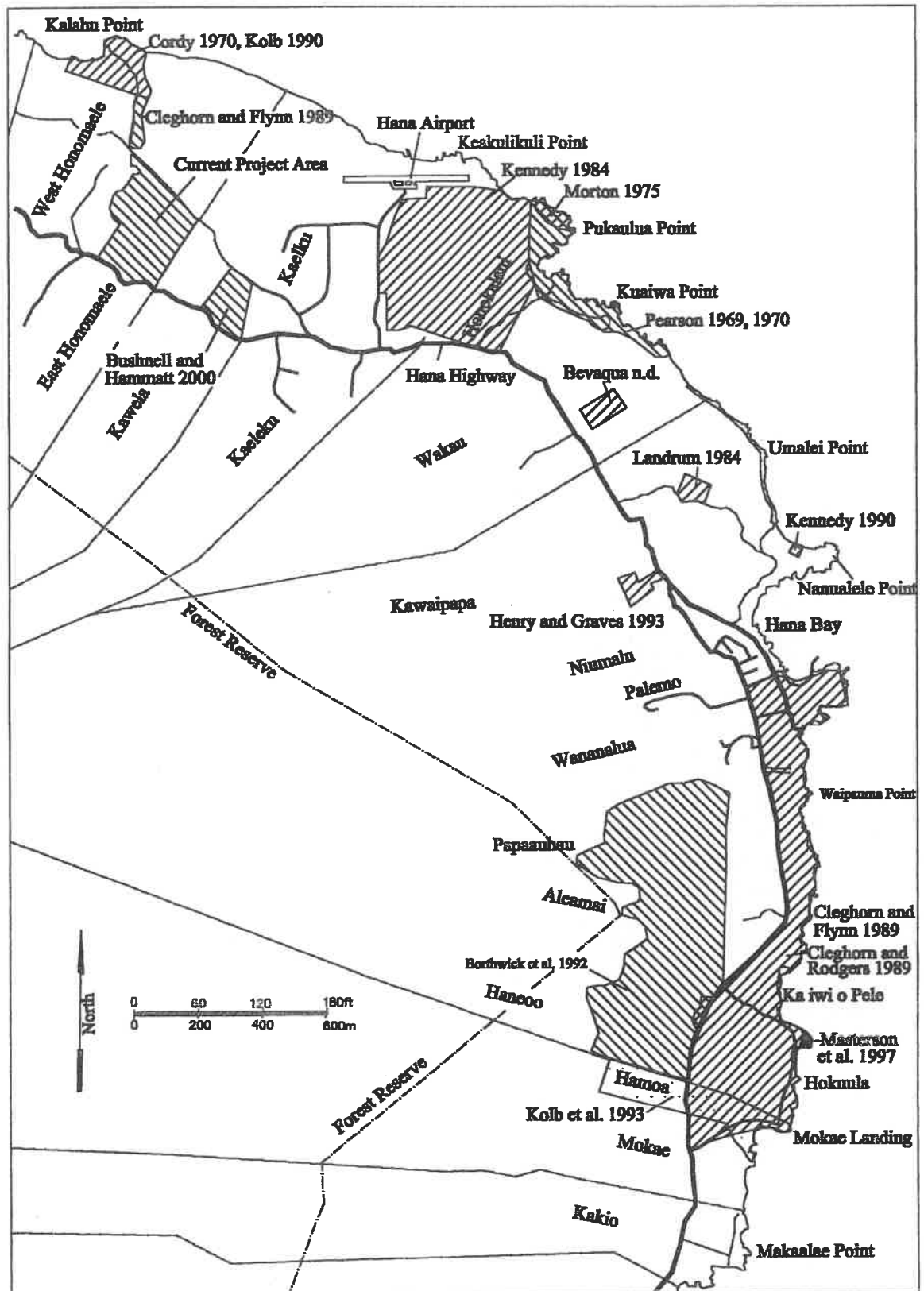


Figure 8. Previous Archaeological Work

Table 1. Summary of Previous Archaeological Surveys

Author	Year	Location	Ahupua'a	Study Type ¹	Area (ac)	Elev. (ft)	Historic Use	No. Sites	Hab Feas	Ag Feas	Burial Feas.	Ritual Feas.	Fish Pond	Hist Feas
Cordy/ Kolb	1970/ 1990	Coastal	Honomaele	IS	9	0-40	Pasture	13	6		3+	22+		2+
Cleghorn & Flynn	1989	Coastal Plain	Honomaele	RS	126	0-70	Cane & Pasture	14	3	24+	5+	1+		1+
Bushnell & Hammett	2000	Lower Slopes	Kaeleku	IS	34	150-350	Cane & Pasture	0						+
Pearson	1970	Coastal	Waikiu	IS	83	0-40	Pasture	5	14	4?	Many	1		4
Morton & Lum Ho/ Kennedy	1975/ 1984	Coastal- Coastal Plain	Waikiu, Kaeleku	RS	364	0-200	Pasture ?	1			Many	1		
Bevacqua	1972	Coastal Plain	Waikiu	RS	16		?	1	1					
Landrum	1984	Coastal	Kawaipapa	RS	14	70-90	?	0						1
Kennedy	1990	Coastal	Kawaipapa	IS	1	0-10	?	1				1		1?
Henry & Graves	1993	Lower Slopes	Kawaipapa	IS	10	160-200	Cane & Pasture	4		1				7
Cleghorn & Rogers (Cleghorn & Flynn)	1987	Coastal, Lower Slopes	Haneoo to Wanahalua	RS	581	0-200	Cane & Pasture	57	17	14+	Many	25	8	11+
Borthwick et al.	1992	Lower Slopes	Haneoo, Aleamai, Papaaupahu Oloewa	IS	400	200-760	Cane & Pasture	51	16	15+	6?	1		42+
Masterson et al.	1997	Coastal	Haneoo	IS	1.5	0-20		5	1		6+	1	2	2
Kolb	1993	Coastal/ Lower Slopes	Hamoia	IS	51+	0-440	Cane & Pasture	18	2	63+	1+	4		1

¹ IS=Inventory Survey, RS=Reconnaissance Survey

the north and Puuiki to the south. Many of the sites have been destroyed. The data indicate that 12 were medium-sized, *mapele*, heiau; six were large, possible *luakini*, heiau; six were shrines, *ko'a*; and two were places of refuge, *puuhonua*. The distribution of these sites is not uniform within Hana District. The majority of heiau, 30 sites, are situated near Hana Town and along the coast to the south of town. Of these, roughly two-thirds are situated on the coast and the remainder are situated at the base of the lower mountain slopes.

The survey reports in *Table 1* cover approximately 1,690 acres of Hana District identifying 161 sites. The survey location is categorized as "Coastal", "Coastal Plain", and "Lower Slopes". The coastal plain is defined here as the broad gently sloping plain between the shoreline and the lower mountain slopes between Ulaino and Hana Town. South of Hana Town the coastal zone borders the lower slopes. To aid in reconstructing settlement patterns, features were quantified by probable age and function. Traditional Hawaiian features were categorized as habitation, agricultural, burial, ritual, and fishpond. Historic features were not segregated by function. Features not clearly assignable to these categories were omitted. The following discussion summarizes the studies beginning in the north and proceeding south.

No previous surveys have been conducted in East Honomale, but several studies have been conducted in the coastal and coastal plain portions of West Honomale (Cordy 1970, Kolb 1990, Cleghorn and Flynn 1989). Cordy (1970) cleared and mapped Piilanihale Heiau and surveyed the surrounding area. The survey identified two house platforms, a house site, three graves, a circular pit, three walls, a complex consisting of a wall, platform and enclosure; and a large enclosure, which formerly contained a number of houses. Other identified features, which were not recorded, consist of a post-1950s house site and a cemetery with at least 14 historic graves. The house site and cemetery are situated on top of the cliff at Kalahu Point. The cemetery and Piilanihale Heiau were also described by Nakim (1970).

Kolb (1990) conducted excavations at Piilanihale Heiau, which he says is the largest heiau in Hawaii. The excavations identified ritual and habitation areas and four major building episodes. Four radiocarbon age ranges span the period between AD 1270 and the mid-1900s. Kolb suggests that the complex may also have served as a chiefly residence.

Cleghorn and Flynn (1989) conducted a survey of Kahanu Gardens, which surrounds the area surveyed by Cordy. The report also describes nine sites recorded on Hana Ranch lands south of Hana Town. At Kahanu Gardens, the survey identified a boulder with cobbles piled on top, a retaining wall, an upright stone, two stone alignments in a stream bank, a low wall, two terraces, a buried stone alignment, a C-shape wall, and four feature complexes. One complex, Site 50-Ma-A10-23, consists of a L-shaped, linear mound, pavement, and overhang associated with hammerstones, cores, and flakes. Site 50-Ma-A10-24 consists of three modified boulders on a rocky beach. One boulder has indentations believed to be an unfinished *papamu*, one has four depressions thought to be bait cups, and the other boulder has a petroglyph of a human form. Site 50-Ma-A10-25 consists of three platforms believed to be graves or a shrine. Site 50-Ma-A10-26 consists of a modified outcrop, a wall, and an enclosure. A hammerstone and basalt core and flake were collected from the site. No other interpretations are offered and no excavations were conducted.

Bushnell and Hammatt (2000) conducted an inventory survey of a 34-acre parcel in Kawela between the Hana Highway and Ulaino Road. The project area ranges in elevation from 150 ft to 300 ft. No sites were identified, although piles of stone were noted throughout the area. The absence of sites and piles of stone are attributed to plantation-era cultivation.

Pearson (1970) conducted a survey of Waianapanapa State Park in Waikiu. The survey covered an area of approximately 83 acres between the coast and approximately 40 ft elevation. The survey identified 34 features that were grouped into five complexes of related features: a heiau and caves, fishing shelters (caves), markers (*ahu*) and a coastal stepping-stone trail, inland permanent house sites and enclosures, and graves or cemeteries. Pearson also identified a pictograph rendered in red ochre. A network of walls enclosing what were reported to be historic house sites was not recorded. No excavations were conducted. The sites were interpreted to be prehistoric to historic period in age.

Kennedy (1984) conducted a survey of approximately 364 acres between the coast and 200 ft elevation in Kawela and Waikiu Ahupua'a. One site, a large complex of burial features, was recorded, which was previously identified by Pearson (1970). The survey identified 364 features consisting of filled crevices, platforms, *ahu*, incomplete graves, and a possible religious structure, a multi-tiered platform with upright stones. No counts for feature types are given and the features are not numbered on the site map. The cemetery is presumed to have

been used between 1600 and the late 1800s. No excavations were conducted. Morton and Lum Ho's (1975) hand-written notes and maps appear to describe the seaward portion of the burial site.

Bevacqua (1972) conducted a survey of approximately 16 acres situated between 40 ft and 100 ft elevation in Waikiu. Only one site, a partially destroyed habitation, was identified. Landrum (1984) conducted a reconnaissance survey of 14-acre parcel in Kawaipapa situated between 70 ft and 90 ft elevation. The only site identified was a segment of the old government road. Kennedy (1990) conducted a reconnaissance survey of an approximately 1.6 acre parcel next to the coast in Kawaipapa. The survey identified Kauleiula Heiau, which was previously described by Walker (1931). A portion of the *heiau* was previously used for a historic house. No excavations were conducted.

Henry and Graves (1993) conducted a survey of a 10-acre parcel situated between 160 ft and 200 ft elevation in Kawaipapa. The survey identified two historic ranch walls and two complexes of features. One complex consisted of two enclosures, an L-shaped alignment, a terrace, and a platform. Excavations were conducted in several features at the site. The excavations produced food remains and historic artifacts indicating a historic habitation use for the site. The other complex consisted of a historic wall and an agricultural terrace.

Cleghorn and Rogers (1987) conducted background research for Hana Ranch lands seaward of the coastal highway between Hana Town and Hamoa. The project area of 581 acres ranges from sea level to approximately 200 ft elevation. Thirty-two sites were identified through background research, examination of aerial photographs, and field inspections. At least 12 of these sites, mostly *heiau* identified by Walker (1931), had been destroyed. A subsequent survey of the "coastal fringe of the Hana Ranch lands" (Cleghorn and Flynn 1989:5) identified nine additional sites. It is unclear from the reports whether the entire 581-acre area was systematically covered. The sites include eight fishponds, at least 25 ritual features, numerous burials, more than 14 agricultural features, and at least 17 habitation features. The habitation features include temporary shelters, primarily in caves, and probable permanent habitations represented by enclosures and platforms. Probable agricultural features consist of terraces, pits, walls, and mounds. Historic features included burials, habitations, and plantation infrastructure. No excavations were conducted.

Borthwick, Robins, Folk, and Hammatt (1992) conducted a survey of 400 acres of Hana Ranch land between approximately 200 ft and 760 ft elevation. The survey identified 51 sites consisting of at least 80 features. Most features consisted of ranch and sugar cane plantation remains including walls, enclosures, platforms, terraces, roads, and a railroad grade. Probable traditional Hawaiian sites included habitations, agricultural features, a *heiau*, and burials. Most traditional sites were described as remnants disturbed by historic activity. Two intact habitation sites were interpreted to be temporary habitations associated with agricultural activity. Probable traditional agricultural features included terraces, pits, walls, enclosures, and mounds. Excavations were conducted at several sites. Three radiocarbon samples produced age ranges of 1345-1650, 1425-1950, and 1640-1950.

Masterson, McDermott, and Hammatt (1997) conducted a survey and subsurface trenching in a 1.5-acre parcel on the coast in Haneo'o. The five recorded sites consist of Haneo'o Fishpond Complex, historic graves, a historic house site, a ranch wall, and a hearth. Excavations yielded food remains and artifacts consisting of both historic and traditional Hawaiian types.

Kolb (1993) reports research conducted in the *ahupua'a* of Hamoa. The research included survey of 51 acres inland of the highway and an unspecified acreage between the highway and the coast. The survey identified 18 sites consisting of more than 70 features. The majority of features were agricultural terraces, walls, and pits assigned an indeterminate "prehistoric/historic" age. Ritual sites consisted of three named *heiau* and a notched enclosure. Habitation features consisted of a cultural deposit in a sand dune and a rectangular enclosure. Excavations at several sites produced seven charcoal samples that yielded age ranges spanning the 1200s to mid-1900s.

Summary of Land Use

Overall, the archaeological surveys conducted in Hana District have identified a relatively small sample of the traditional Hawaiian sites that were formerly present. The massive impacts of sugar cane cultivation and ranch-related pasture improvement and infrastructure have obliterated much of the pre-contact cultural landscape. Numerous *heiau*, burial sites, and fishponds along the coast attest to the presence of a substantial pre-

contact population. Radiocarbon dating results indicate settlement by at least the 1200s with most results post-dating the mid-1400s. The first and largest building episode of Piilanihale Heiau in West Honomale dates to between 1270 and 1440 and indicates the presence of a substantial supporting population.

Habitation sites, both temporary and permanent, are present along the coast. Temporary habitations consist of caves, overhangs, and simple walled structures. Permanent habitations are represented by walled enclosures and platforms. Inland habitation sites on the lower mountain slopes are primarily temporary occupations, probably associated with agricultural activity.

Agricultural sites consist of terraces, walls, mounds, pits, and alignments. Typically these features are only found in rocky areas that were not affected by sugar cane cultivation. The agricultural features represent a pattern of informal agricultural plots and not formal fields. Opportunistically placed, informal plots are typical in agriculturally marginal, rocky areas elsewhere in Hawaii. The absence of formal fields may be a bias resulting from historic modification of the more productive areas. Alternatively, conditions may not have required or resulted in the development of formal fields bounded by walls and terraces. The ample rainfall and soil of the district made agricultural use readily productive. Historic accounts attest to the bounty of agricultural produce, primarily grown without irrigation. Cultigens included breadfruit, yams, taro, sweet potatoes, *olona*, *wauke*, *'awa*, and bananas. Upland areas above 1,000 ft elevation were cultivated when seasonal droughts affected the lowlands.

The distribution of heiau and fishponds along Hana's coast between Keaa Beach and Puuiki shows a marked increase in density from Hana Town south. The northern coast from Hana Town to Ulaino has relatively few *heiau*. The area differs environmentally from the southern coast. It is characterized by a broad coastal plain derived from relatively recent lava flows. Unlike the south coast, the coastal settlements are separated from the lower mountain slopes by a broad gently sloping plain up to 6,000 m in width. There are only three major drainages crossing the plain at Ulaino, Honomale and Kawaipapa. If the better watered lower mountain slopes were the most productive agricultural area, then the greater distance from the coast may have made the northern coast a comparatively less favored area of occupation. LCA claims appear to support a difference between the north and south. The Waihona 'Aina database (Waihona 'Aina Corp. 1998) lists 11 LCA claims (6 awarded) in the nine *ahupua'a* from Nahiku to Kawaipapa. There were 72 claims (42 awarded) in the twelve *ahupua'a* from Niunalu to Puuiki.

Legendary and traditional accounts document the importance of Hana District as a seat of social and political power, especially in relations between the chiefs of Maui and Hawaii Islands. This prominence continued into early historic times. Historic habitations and burial sites were scattered along the coast. Small areas of subsistence agriculture apparently continued in use into the 1900s; however, for nearly 100 years between the 1840s and 1940s sugar cane cultivation was the dominant form of land use. Nearly all readily cultivated areas of the lower mountain slopes and coastal plain were put into production by up to six plantations. Cartographic and documentary evidence illustrate the aggressive acquisition of land for cultivation and development of plantation infrastructure. Roads, flumes, and a railroad system were developed by the plantations. Mill operations, harbor facilities, and a series of laborer camps were established.

LCA testimony indicates sugar cane was cultivated in Honomale by 1846. By 1851, Kelk and Needham, one of the principals in the original Hana Plantation, owned over 400 acres in East Honomale and undoubtedly used it for sugar cane cultivation. From at least the 1870s, August Unna, another principal in the plantation, controlled the remainder of East Honomale, and large portions of Kawela and Kaeleku through lease with the government. In addition, he apparently owned substantial acreage. Unna was responsible for the establishing the plantation railroad system in the area by 1883.

The original government road, Okaka Puu Road, in Kaeleku, Kawela, and Honomale followed the route of today's Ulaino Road. The upper Government Road, today's Hana Highway west of the junction with Ulaino Road, was constructed between 1894 and 1900. By 1915, a small settlement was present at the junction of the roads. The first railroad tracks in East Honomale are shown on a map dating to 1915. This spur was shortened by the 1920s when it terminated to the east in Kaeleku. A second railroad spur extended along the base of the slopes to a small settlement or camp in West Honomale. A flume paralleling Honomale Gulch in East Honomale was constructed between 1915 and the 1920s. In the waning years of the plantation, cultivation was focused in areas closest to the transportation system (McCall 1940). East Honomale was probably cultivated until the end of the plantation in the 1940s because of its established transportation infrastructure.

After 1945, the former sugar cane lands were converted to pasture for Hana Ranch. Pasture was improved using bulldozers beginning in the 1960s. Additional bulldozing has occurred since the 1980s.

PROJECT EXPECTATIONS

Prehistoric use of the project area is potentially evidenced by relatively rare, scattered temporary habitation sites dating to as early as the 1200s. Such sites should become more common after the mid-1400s until the early historic period. Temporary habitation sites (caves and small walled shelters), trails, and agricultural features, such as terraces, terraces, enclosures, pits, and mounds, were probably present.

Sites dating to the mid- to late 1800s would include a few scattered examples of the agricultural and habitation sites mentioned above. Sugar cane plantation-related sites should dominate the landscape, primarily represented by transportation infrastructure and piles of stone resulting from field clearing. By the 1940s, traditional agricultural and habitation sites should be rare. Ranching activity, which continues today, would be evidenced by walls, corrals, and clearing piles of stone associated with pasture improvement.

FINDINGS

The archaeological survey identified four sites with seven component features. The sites consist of two complexes of railroad features associated with the transportation of sugar cane (Sites 4963 and 4964), a road (Site 4965) and a human burial (Site 4966). The identified sites and features are discussed below and their locations are illustrated in *Figure 9*.

Site 4963

Site 4963 is a complex of three features located in the *makai* portion of the project area (see *Figure 9*). These features consist of concrete and stone abutments and piers that supported plantation railroad bridges. Historic maps, presented in the Historic Documentary Research section of this report, indicate that these bridges were constructed between 1915 and 1923. No evidence of a railroad grade was found extending between the features. This grade has apparently been destroyed by the extensive modern and historic agricultural and ranching use of the area.

Feature A

Feature A is a railroad bridge abutment and stream bank retaining wall located within the *makai* drainage in the northeastern portion of the project area (*Figure 10*). The feature is situated at c. 90 ft elevation in an area of relatively level terrain. The bridge abutment is comprised of two formed concrete piers; one located in the southeastern bank of the streambed, and the other situated within the center of the drainage. Both walls are oriented in a northeast-southwest direction. The first pier is c. 4.5 m long, 0.48 m wide and 0.17 m in height above the stream bank. The second pier is situated c. 2.8 m to the northwest of the first. It is c. 3.4 m long, 0.35 m wide and 0.95 m in height (*Figure 11*). There are two, 0.09 m in diameter, metal eyebolts located on each side this side of the pier.

The retaining wall is located on the northern side of the drainage. The eastern portion of this wall is stacked stone that has been coated in concrete. It has an overall length of 15.8 m, and is 0.75 m wide and ranges in height from 1.55 to 2.0 m above the floor of the drainage. The top of the wall is level with top of the stream bank. The eastern end of this wall is located at the bend in stream. The western section of the retaining wall is built of stacked basalt boulders and cobbles (*Figure 12*). It is c. 12.6 m long, 0.75 to 0.85 m wide and 0.8 to 0.95 m in height. No cultural remains were found in association with this site.

Feature B

Feature B consists of two parallel concrete piers located within a small streambed near the northern project area boundary. The feature is situated at the c. 80 elevation in a level area that slopes slightly to the north-northwest. The piers are constructed of formed concrete. No cultural remains were present.

The first pier is built into the western bank of the stream. It is c. 3.2 m long (NNE-SSW), 0.5 m wide, and 0.95 m tall above the streambed (*Figure 13*). The top of the wall is level with the surrounding ground surface. The second pier is located within the streambed, c. 3.95 m east of the first. The pier is c. 3.5 m long (NNE-SSW), 0.38 m wide, and 0.99 m in height above the streambed. The southern, upstream end of the wall is rounded, to reduce the effect of strong currents during periods of heavy water flow.

Feature C

Feature C consists of two stone and mortar bridge abutments located on the north and south banks of Honomaele Gulch (*Figure 14*). The feature is situated in the northern corner of the project area at c. 70 ft elevation. The terrain in the area is relatively level, sloping slightly to the north-northeast. The sides of the stream gulch consist of vertical basalt bedrock walls. The abutment on the northern bank is comprised of a vertical stone wall. The wall is made from well-fitted and mortared subangular basalt cobbles and small boulders (*Figure 15*). The wall is c. 5.0 m long (NNE-SSW), 0.7 m wide, and from 2.55 to 2.7 m in height above the streambed. The upper surface of the wall is capped with concrete.

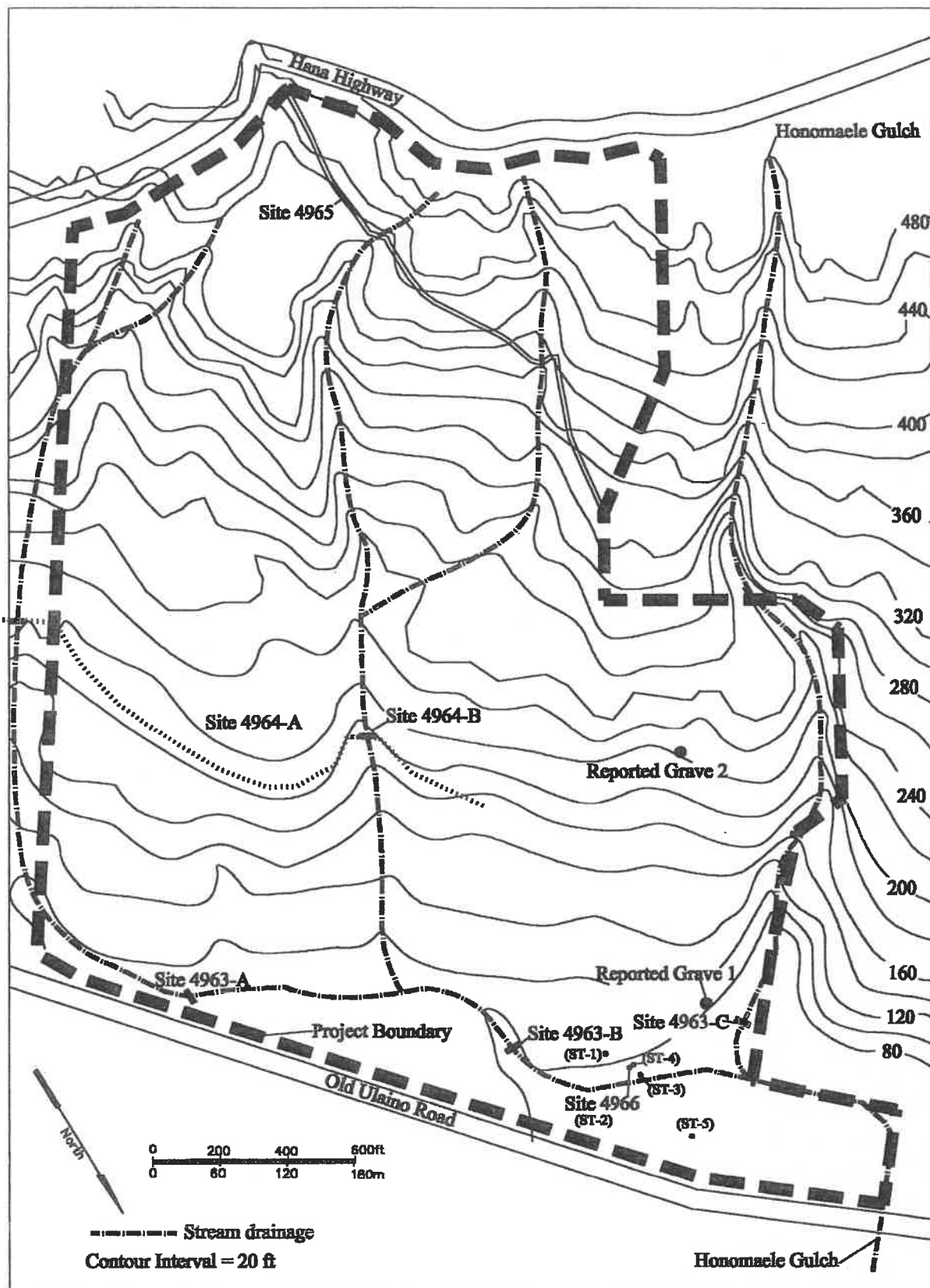


Figure 9. Site Location Map

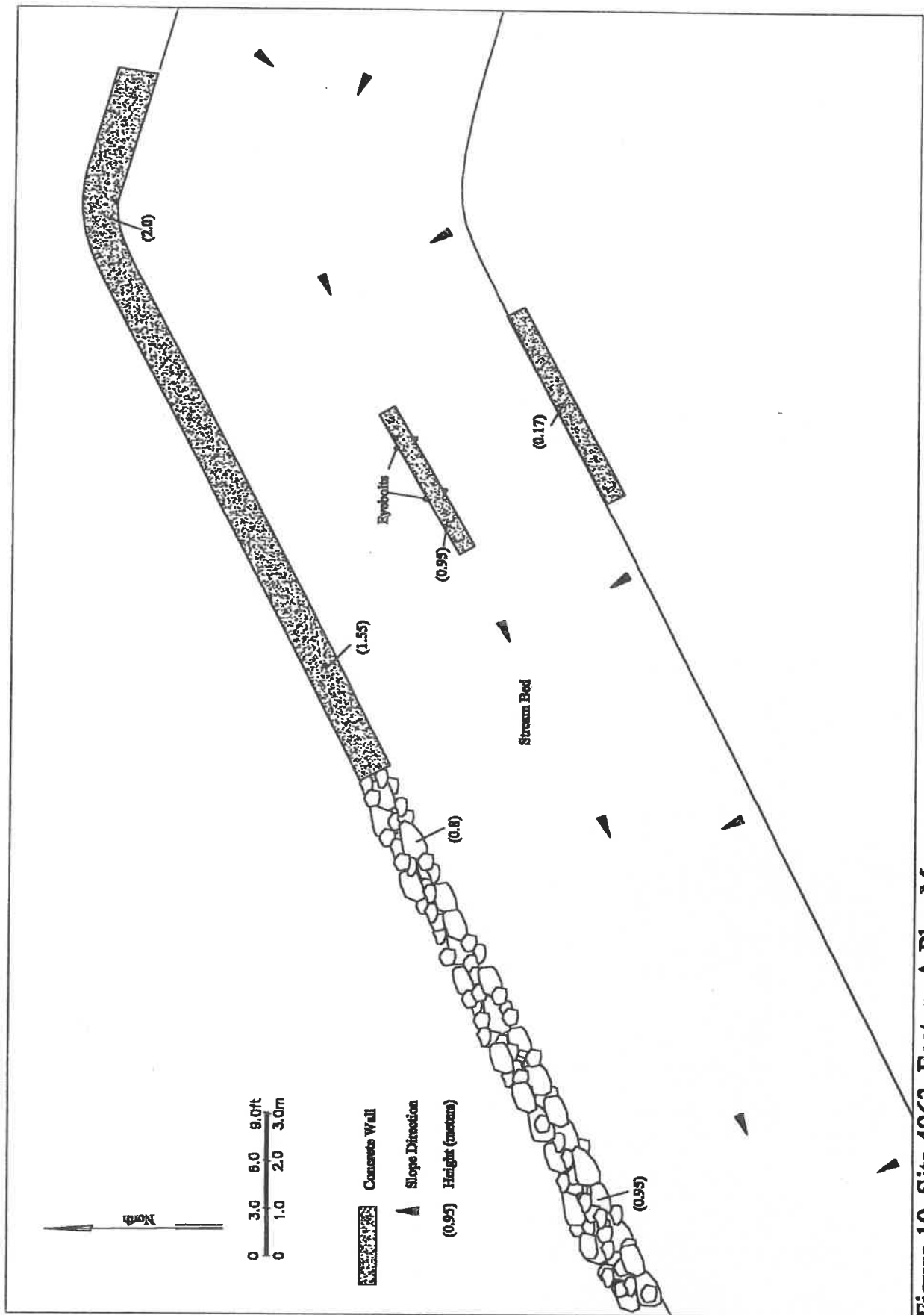


Figure 10. Site 4963, Feature A Plan Map



Figure 11 - Site 4963, Feature A, view to southeast



Figure 12. Site 4963, Feature A, view to northwest

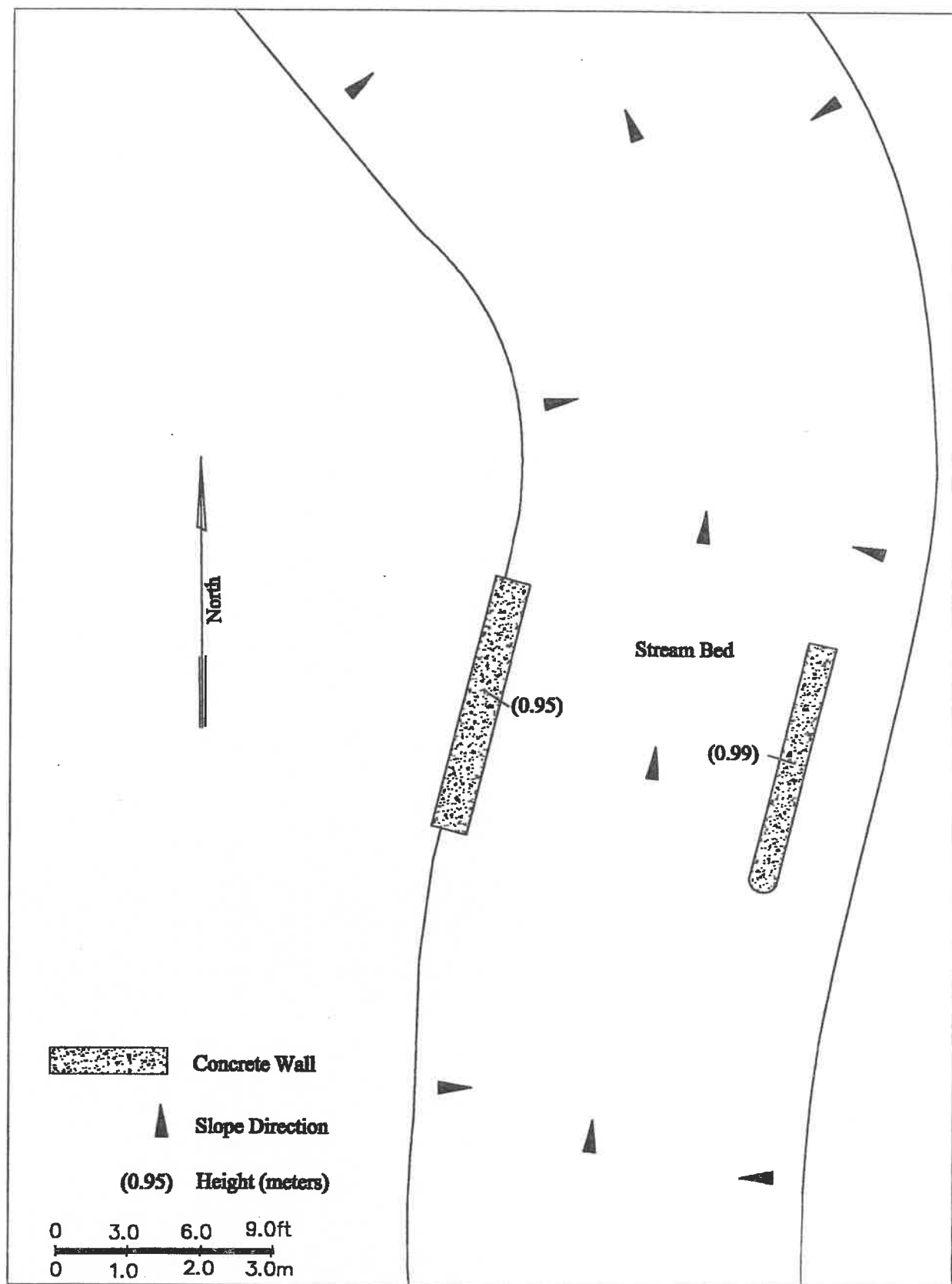


Figure 13. Site 4963, Feature B Plan Map

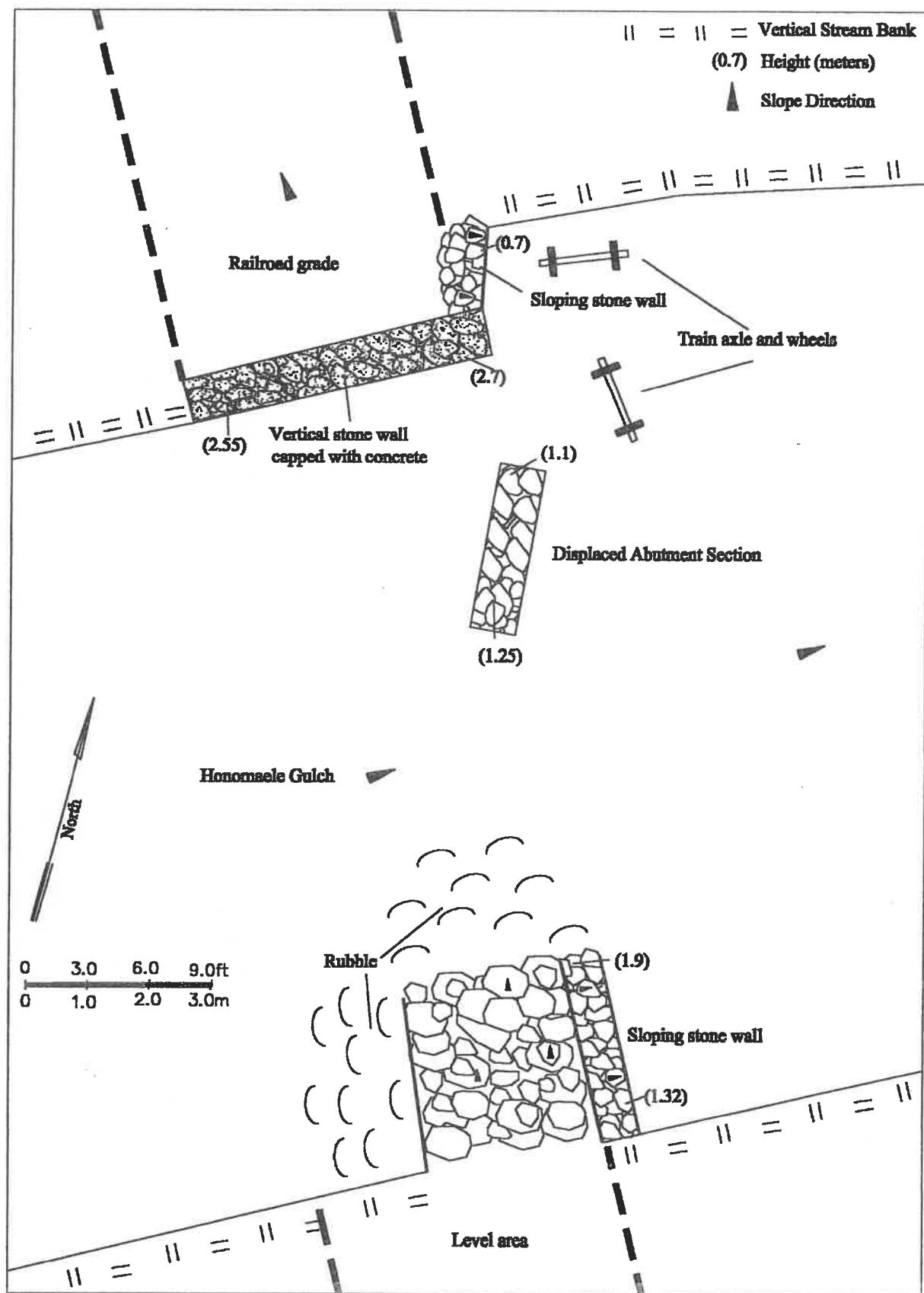


Figure 14. Site 4963, Feature C Plan Map

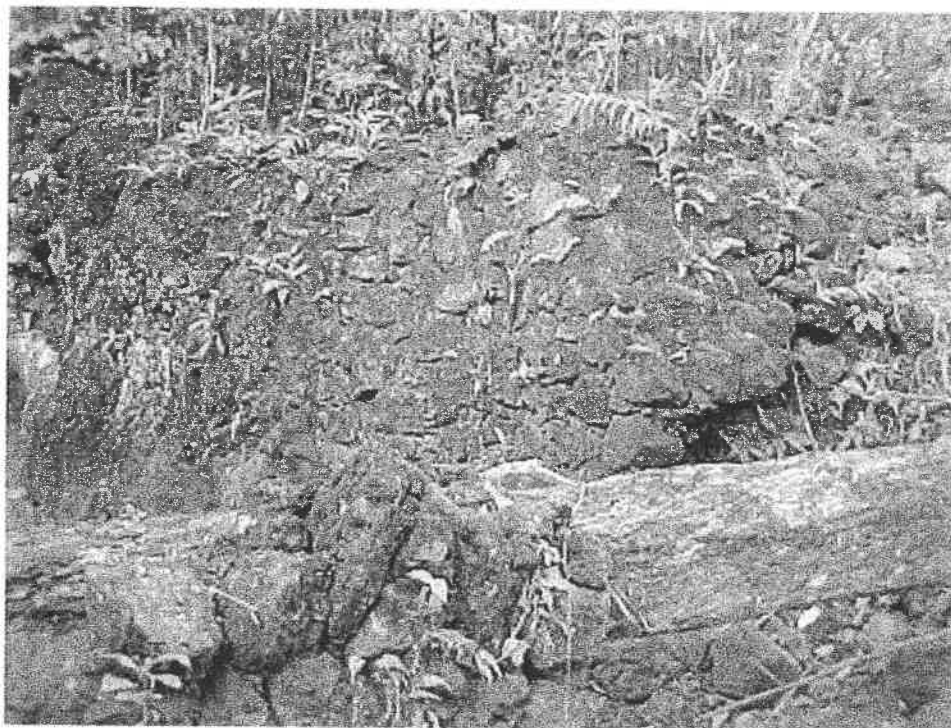


Figure 15 - Site 4963, Feature C, north abutment, view to north



Figure 16. Site 4963, Feature C, railroad car wheels and axle, view to north

The western end of the wall abuts the vertical bedrock bank of the drainage. A second wall built of stacked stones extends from the eastern end of the main wall to the north-northwest 1.4 m to the stream bank. This wall slopes upward from the base to the top, measuring c. 0.7 m in height at the northern end.

The area on top of the abutment is level and appears to be paved with gravel, representing the railroad grade, which extends to the north-northwest out of the project area. No rails were present. Two axles and wheels from a train car are situated adjacent to the northern abutment. The axles measure 1.44 m in length and 0.08 m in diameter. The wheels are 0.46 m in diameter and 0.11 m thick (*Figure 16*).

The abutment on the southern side of the drainage is in poor condition. It consists of a pile of basalt boulders, which slopes down into the drainage to the north-northwest. This boulder pile is 3.0 m long (NNW-SSE) by 2.6 m wide. It probably represents the original base of the abutment. The southern end is level with the top of the stream bank and the northern end is c. 1.7 m above the streambed.

There is an intact, stacked stone wall on the eastern side of the abutment. This sloping wall extends from the vertical stream bank to the north-northwest 3.0 m. It is 0.75 m wide and ranges in height from 1.32 to 1.9 m. It is constructed of fitted cobbles and boulders that have been mortared together. There is a section of mortared stone wall located within the drainage that probably was displaced from this abutment. It is c.2.7 m long (north-south), 0.75 m wide and from 1.1 to 1.25 m tall. This section of wall may have once formed the northern edge of the southern abutment.

Site 4964

Site 4964 consists of a railroad grade (Feature A) and a stone bridge abutment (Feature B) located on the side of a drainage (see *Figure 9*). The site is situated at elevations ranging from 155 to 205 ft. The site is in fair condition and appears unaltered.

Feature A

Feature A originates outside the eastern project area boundary. There is a stone bridge abutment located within the drainage in this area. It is situated outside the project area and was not recorded. The railroad grade extends to the west-northwest from this abutment a distance of c. 35.0 m, angles to the northeast for 32.0 m, and turns again to the west-northwest. It continues in this direction for c. 275.0 m where it enters the *mauka-makai* drainage in the center of the project area. This section of Feature A terminates at the southeast side of the drainage at the Feature B abutment. The grade continues on the northwest side of the drainage in a roughly north-northwesterly direction for c. 115.0 m. Beyond this point, no trace of the railroad grade visible.

The railroad grade is comprised of a crushed gravel and soil bed that varies in width from 3.5 to 4.0 m. In some places, the *mauka* side of the grade has been excavated into the slope. The *makai* side has been built up with soil. Metal fragments are present on the surface.

Feature B

Feature B is the stone bridge abutment located on the southeast side of the drainage (*Figure 17*). The feature is comprised of a vertical stone wall that faces the drainage. This wall is constructed of tightly fitted subangular basalt cobbles and boulders with no evidence of mortar. The edge of the abutment wall is c. 3.9 m wide and 5.1 m tall above the streambed. The sides of the abutment extend from the northwestern edge back towards the top a vertical bedrock face. The walls slope up from the drainage floor to the top of the railroad grade. These sides are also constructed of fitted basalt stones. The walls average 2.1 m wide and 2.5 to 5.8 m in height. *Figure 18* illustrates Feature B.

There is a linear stone mound that extends to the northeast from the edge of the abutment. This mound is built of piled basalt boulders and cobbles, and is c. 3.7 m long, 2.4 m wide, and 1.0 m in height. This wall probably constructed to divert the flow of water away from the base of the abutment. There are two vertical metal bars associated with Feature B. These bars are 0.055 m by 0.045 m in size. One bar is

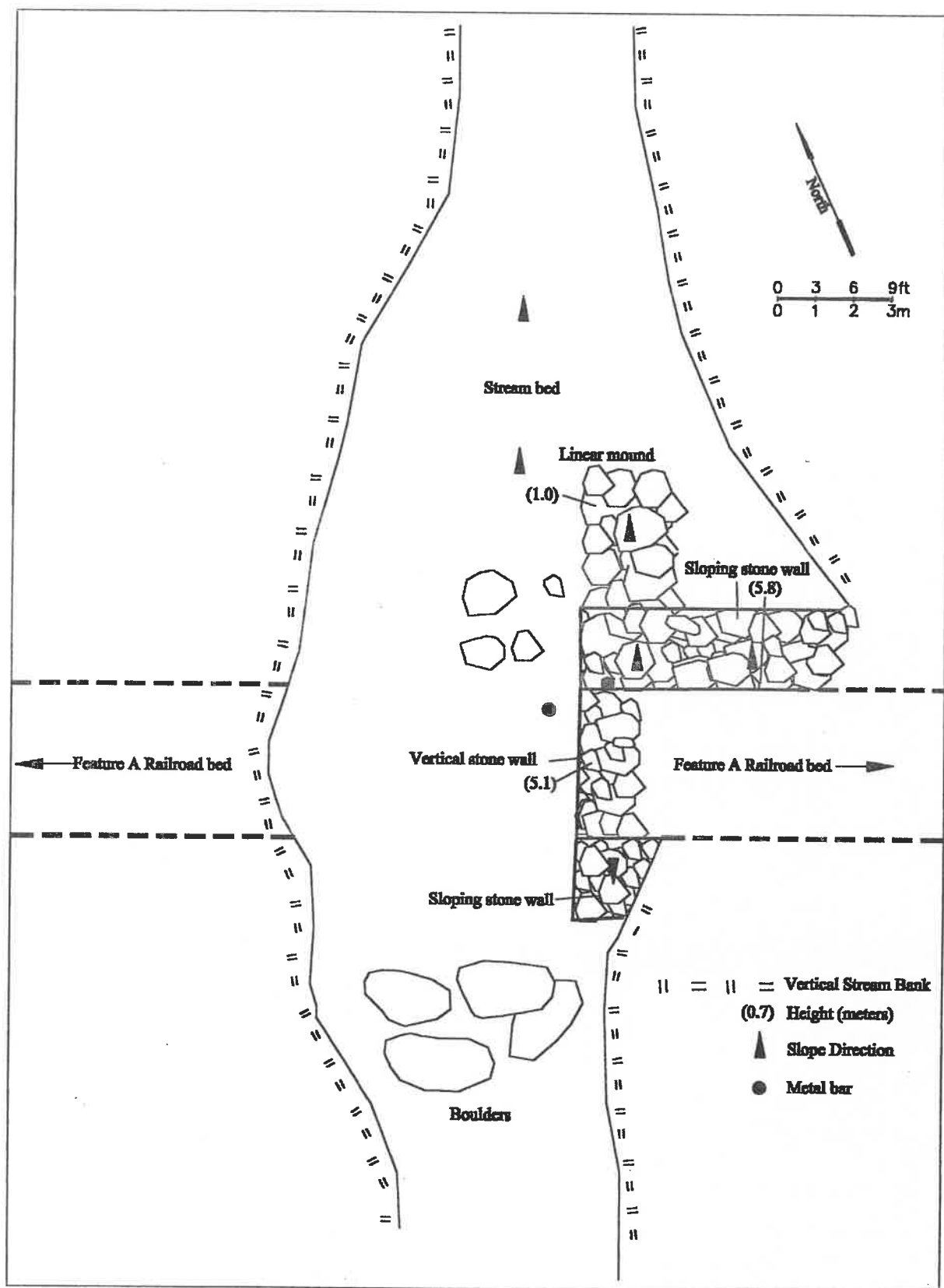


Figure 17. Site 4964, Feature B Plan Map



Figure 18 - Site 4964, Feature B, view to south



Figure 19. Site 4965, view to south

incorporated into the top of the abutment at the northern corner, measuring c.0.25 m tall. The second bar is located within the drainage, c. 0.9 m northwest of the abutment. This bar is c. 3.5 m in height. These bars were probably structural components of the bridge.

There is no abutment on the northwestern side of the drainage. This side of the streambed consists of a vertical stone face that is c. 2.9 m in height. Remnants of the Feature A railroad grade are present adjacent to the vertical face.

Site 4965

Site 4965 is a roadbed located in the *mauka* portion of the project area at elevations ranging from 330 to 480 ft. The road extends in a roughly north-south direction across the slope, which angles moderately to the north-northeast. The road ranges in width from 2.25 to 3.0 m with a surface of gravel and soil (Figure 19). The overall length of the road is c. 480.0 m. There are cut banks on the upslope side of the road. The cut-banks range in height from 0.4 to 1.4 m. Corresponding retaining walls are located on the downslope side of the road, built of piled subangular basalt cobbles and small boulders and soil. The retaining walls range in height from 0.2 to 0.45 m. No cultural remains were found in association with this road. This site is unaltered and in fair condition. The road probably post-dates the 1920s because it does not appear on any early maps. It was probably constructed and used in conjunction with sugar cane cultivation or ranching activity.

Site 4966

Site 4966 is a human burial located in the northwestern portion of the project area at c. 80 ft elevation. The remains were identified during testing for prehistoric pondfield deposits in the level floodplain along the northern project area boundary. The skeleton was encountered in ST-2. This unit originated as a 0.5 m diameter shovel test unit. The excavation revealed a 0.3 m thick layer of dark brown gravelly silt clay loam (Layer I) above a dark brown layer of very gravelly (cinders) silt clay (Layer II). Skeletal material was encountered within the lower deposit at 0.49 to 0.53 m below surface. Two additional Shovel Tests (ST-3 and -4) were excavated in the immediate vicinity of the burial with negative results.

Upon identification of potential human remains, the shovel test was expanded to a 0.75 by 0.5 m unit. The careful excavation of the remains revealed two articulated ribs. Small rib fragments, a phalanx, carpal, and possible vertebra fragment were recovered from the screened fill. A basalt flake and charcoal were found in association with the remains. No evidence of a pit feature was identified. The charcoal was submitted for radiometric age determination, yielding four potential calibrated calendric age ranges of AD 1470-1700, 1720-1820, 1835-1880, and 1915-1950 (Appendix A). Once it was determined that articulation was present, a layer of cotton cloth was placed over the remains and the unit was carefully backfilled. The recovered bone fragments were subsequently examined by consulting osteologist, Dr. Diane Trembly, who confirmed the bones were human.

Subsurface Testing

As stated, subsurface testing was undertaken in the *makai* portion of the project area to determine if evidence of prehistoric pondfield cultivation was present. The tested area is a level plain bisected by a small streambed, which drains the area in a west-northwest direction. Five shovel tests were excavated in this area (see Figure 9). Figure 20 presents a schematic view of the soil stratigraphy identified during testing.

The surface layer throughout the tested area was comprised of a culturally sterile deposit of silty clay loam (Layer I) that varied in thickness from 0.12 to 0.5 m and averaged 0.25 m. The deposit is an alluvial deposit. The surface deposit is underlain by a layer of pea-sized and smaller black cinders (Layer II). The cinders are mixed with silty clay in ST-2. The layer potentially represents an eruptive episode of a nearby cinder cone. Cultural remains were present at the base of this layer in two of the shovel tests. These include the human skeleton, charcoal, and a basalt flake identified in ST-2. A single basalt flake was recovered from ST-4.

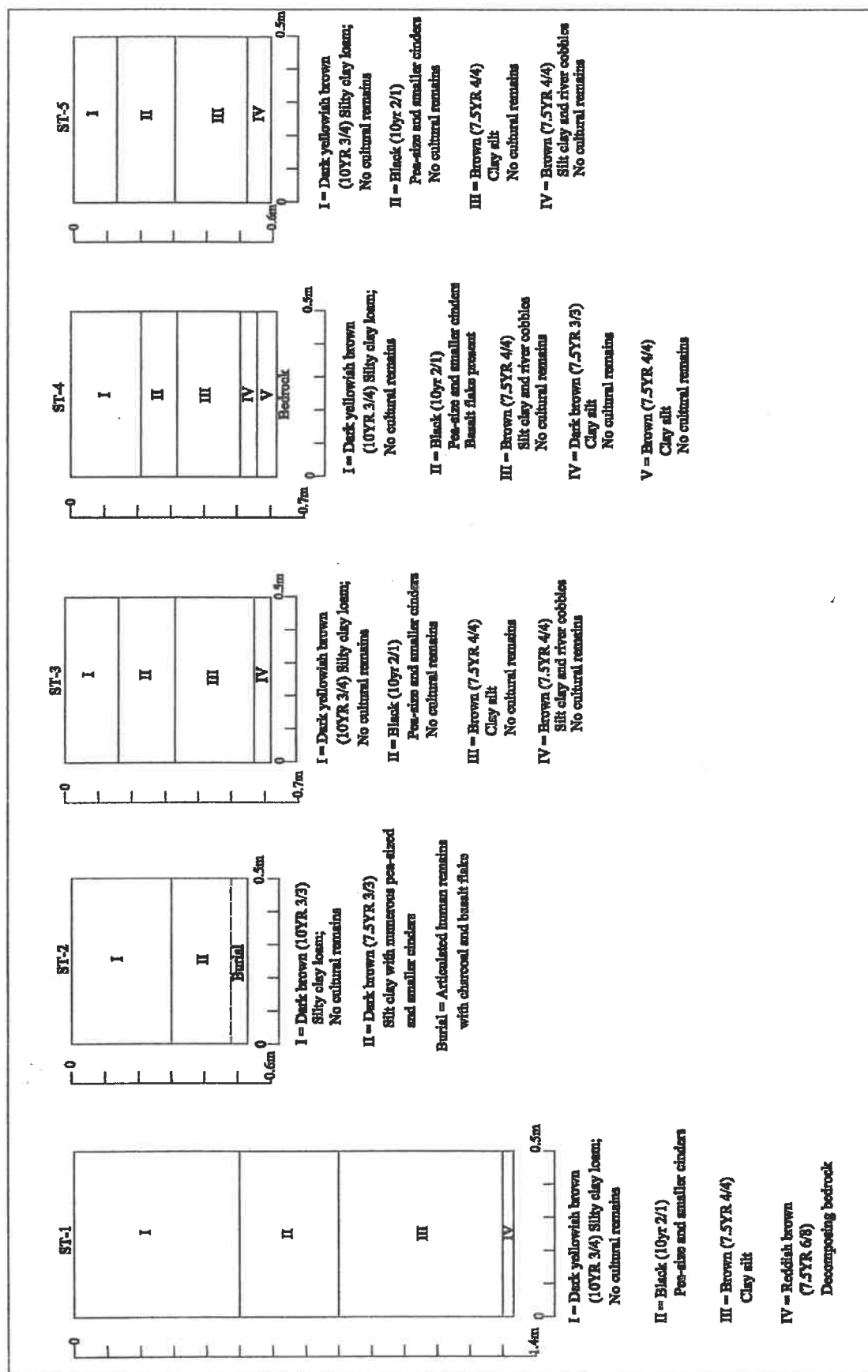


Figure 20. Shovel Test Stratigraphy

Layer III is a layer of brown clay silt encountered in four shovel test units (ST-2 was terminated when human remains were identified). In STs 3 and 5, Layer III was underlain by a brown silt clay mixed with small waterworn river cobbles (Layer IV). In ST-1, this lower soil was comprised of a reddish brown decomposing bedrock (Layer IV). The excavation of these three STs was terminated in Layer IV. In ST-4, two soil deposits were present below Layer III. Layer IV consisted of a dark brown clay silt and Layer V was a brown clay silt overlying waterworn bedrock.

Reported Graves

Two locations within the project area were reported to contain historic human burials (see *Figure 9*). According to Harry Koon, an employee of the present owner and resident of Hana for over 20 years, a former employee of Hana Ranch, a c. 70 year-old man named "Blue", told him that there were graves in two locations in the project area. Mr. Koon also stated that other former ranch employees indicated that the graves were destroyed by bulldozing activity. One grave site was reportedly located beneath a large tree situated in the northwestern portion of the project area at c. 80 ft elevation. The other was located adjacent to a lone coconut tree, c. 230 m upslope to the south-southwest, at 200 ft elevation. Both areas were carefully examined during the current study. No evidence of graves was present. The areas have been mechanically disturbed (metal-scarred stones) and are dominated by secondary vegetation.

CONCLUSION

Discussion

The identified sites conform to the types expected based on previous archaeological work and historic documentary research. As expected, historic agriculture and ranching have extensively modified the landscape and destroyed all surface evidence of traditional Hawaiian occupation. The discovery of an intact human skeleton (Site 4966) in the floodplain deposits in the northwestern corner of the project area was not expected. Traditional Hawaiian burials are typically found in coastal sand deposits, lava tubes, caves, and crevices. Burial also occurred in *heiau*, beneath house floors, and beneath stone platforms and cairns.

Floodplains usually were not used for burial or habitation. The stream course, which meanders through the seaward portion of the project area inland of Ulaino Road, has shifted across the floodplain over time. This shifting is evidenced by the waterworn bedrock surface encountered in ST-4 and a deposit of waterworn cobbles encountered in ST-3 and ST-5. The long retaining wall at Feature A of Site 4963 was apparently built to keep the stream from eroding and shifting seaward at its closest point to Ulaino Road. Periodic flooding of the area is well known to local residents. Following prolonged periods of heavy rain, several feet of water cover a large area extending from the northwest corner of the project area to Kahanu Gardens. These floods occur at least once a year.

The skeletal remains were in the lower portion of a deposit containing cinders. The cinders probably were deposited during an eruptive episode of a nearby cinder cone. Although no pit feature containing the remains was evident, the presence of charcoal fragments and a basalt flake suggest that the remains were deposited as a result of cultural activity, and not a natural event such as a flood. The bones are very friable indicating they have been buried for a long period of time. Radiocarbon dating of the charcoal fragments produced four potential age ranges of AD 1470-1700, 1720-1820, 1835-1880, and 1915-1950. The basalt flake indicates that the remains are probably prehistoric to early historic period in age because stone tools were replaced shortly after contact. Based on the condition of the bones and presence of a basalt flake, one of the two early radiocarbon ages probably dates the burial. Two additional test excavations next to the remains did not encounter other features or evidence of a cultural deposit. The remains probably represent an isolated, late prehistoric to early historic Hawaiian burial based on the available evidence.

As expected, the remains of sugar cane plantation transportation infrastructure were identified. The railroad bed and bridge abutment of Site 4964 were constructed before 1915 when the railroad track first appears on a map (see *Figure 6*). The railroad was a spur that terminated in the sugar cane field in East Honomalee at approximately 200 ft elevation. The spur would have been used to transport harvested cane to the mill. August Unna put the railroad system into operation in 1883; however, maps from 1893 and 1900 (see *Figures 4 and 5*) do not show a railroad. Thus, the tracks were probably built between 1900 and 1915. By the mid-1920s, the portion of the spur in the project area appears to have been abandoned. The USCGS map (see *Figure 7*), which is based on surveys between 1923 and 1925, shows the spur terminating at a small settlement in Kawela east of the project area.

By the 1920s, the cane from the fields in East Honomalee was transported by a flume shown on the USCGS map (see *Figure 7*) and a railroad spur (Site 4963), which extended along the seaward edge of the project area. No evidence of the flume is present today. The remains of the railroad spur consist of three stream crossing were bridge abutments and support piers are present. The spur was constructed between 1915 and 1923. The stone abutments in Honomalee Gulch (Feature C) resemble the Site 4964 abutment except the stones are mortared, potentially the result of post-construction maintenance or repair. The piers of Features A and B are formed concrete probably indicating the piers are a later construction. The piers may have replaced stacked stone bridge abutments that were damaged by flooding along the stream. This is suggested by the abutment and retaining wall on the seaward side of the stream at Feature A. The wall is a stacked stone covered with a veneer of concrete. A portion of the retaining wall is stacked stone lacking mortar.

The Site 4965 roadbed does not appear on any maps. The road may formerly have provided access between Ulaino Road and Hana Highway. The road was probably constructed after the 1920s. It may have been constructed as late as the 1960s when bulldozers were being used to improve pasture.

Significance Assessments and Recommended Treatments

Pursuant to DLNR (1998) Chapter 275-6 (d), the initial significance assessments provided herein are not final until concurrence from the DLNR has been obtained. Sites identified and relocated during the survey are assessed for significance based on the criteria outlined in the Rules Governing Procedures for Historic Preservation Review (DLNR 1998:Chap 275). According to these rules, a site must possess integrity of location, design, setting, materials, workmanship, feeling, and association and shall meet one or more of the following criteria:

1. Criterion "a". Be associated with events that have made an important contribution to the broad patterns of our history;
2. Criterion "b". Be associated with the lives of persons important in our past;
3. Criterion "c". Embody the distinctive characteristics of a type, period, or method of construction; represent the work of a master; or possess high artistic value;
4. Criterion "d". Have yielded, or is likely to yield, information important for research on prehistory or history; and
5. Criterion "e". Have an important traditional cultural value to the native Hawaiian people or to another ethnic group of the state due to associations with traditional cultural practices once carried out, or still carried out, at the property or due to associations with traditional beliefs, events or oral accounts--these associations being important to the group's history and cultural identity.

Based on the above criteria, Sites 4963 and 4965 are assessed as significant under Criterion "d". The sites have yielded information important for understanding historic land use in project area. The mapping, written description, and photography at the sites adequately document them and no further work or preservation is recommended.

Site 4964 is assessed as significant under Criterion "d" and Criterion "c". The site has yielded information important for understanding historic land use. Feature B of the site, a dry-laid masonry bridge abutment, is a well-preserved example of plantation railroad architecture dating to before 1915. The feature is recommended for preservation. The specific plans for preservation of the site would be detailed in a Preservation Plan prepared for DLNR-HPD review and approval.

Site 4966 is assessed as significant under Criterion "d" and Criterion "e". The site has yielded, and has the potential to yield, information important for understanding prehistoric mortuary practices and demography. The site is culturally significant because it consists of a pre-contact burial. The site is recommended for preservation. The specific plans for preservation and maintenance of the site would be detailed in a Burial Treatment Plan prepared for DLNR-HPD and the Maui County Burial Council review and approval.

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APPENDIX A – RADIOCARBON CALIBRATION

CALIBRATION OF RADIOCARBON AGE TO CALENDAR YEARS

(Variables: C13/C12=-25.3;lab. mult=1)

Laboratory number: Beta-143510

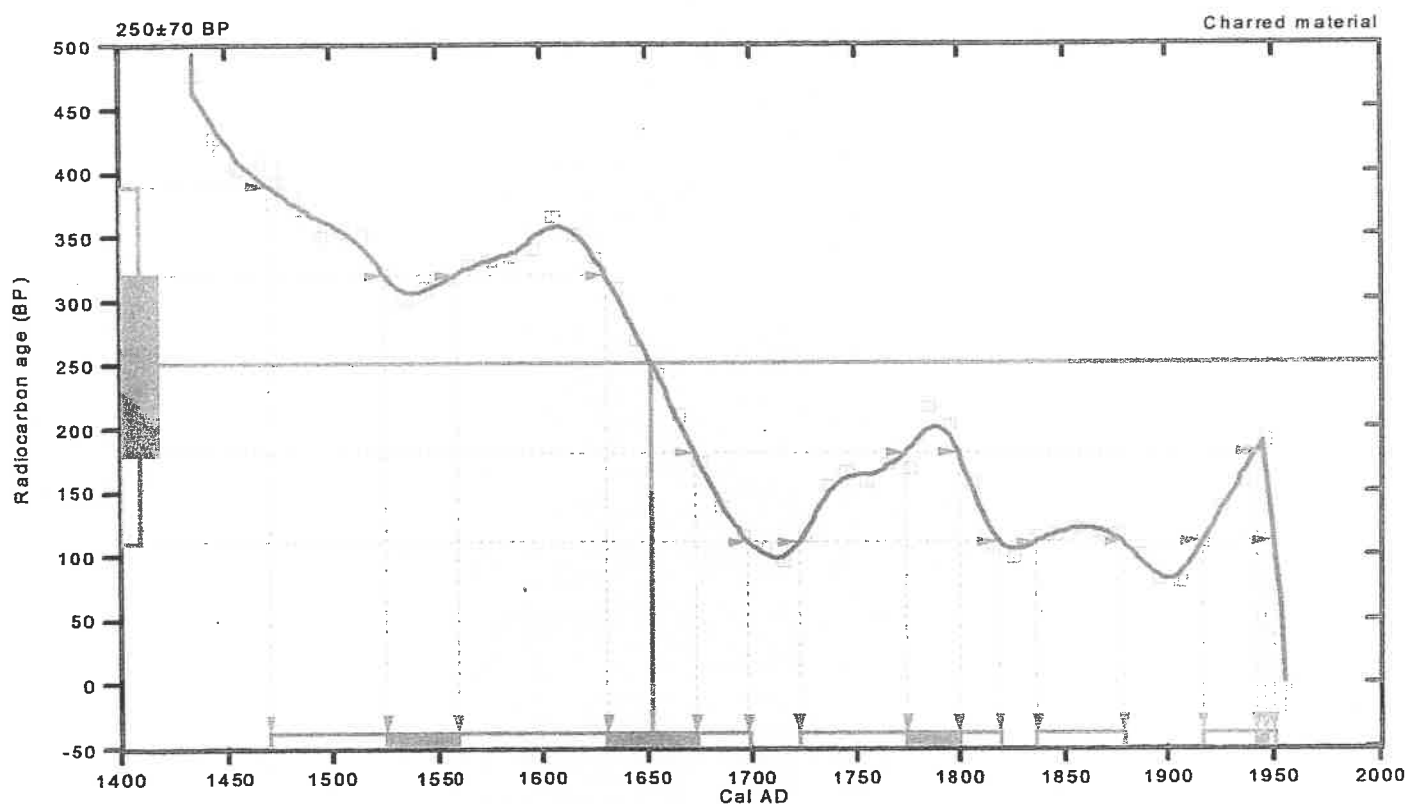
Conventional radiocarbon age: 250 ± 70 BP

2 Sigma calibrated results: Cal AD 1470 to 1700 (Cal BP 480 to 250) and
(95% probability) Cal AD 1720 to 1820 (Cal BP 230 to 130) and
Cal AD 1835 to 1880 (Cal BP 115 to 70) and
Cal AD 1915 to 1950 (Cal BP 35 to 0)

Intercept data

Intercept of radiocarbon age
with calibration curve: Cal AD 1650 (Cal BP 300)

1 Sigma calibrated results: Cal AD 1525 to 1560 (Cal BP 425 to 390) and
(68% probability) Cal AD 1630 to 1675 (Cal BP 320 to 275) and
Cal AD 1775 to 1800 (Cal BP 175 to 150) and
Cal AD 1940 to 1945 (Cal BP 10 to 5)



References:

Database used

INTCAL98

Calibration Database

Editorial Comment

Stuiver, M., van der Plicht, H., 1998, Radiocarbon 40(3), pxi-xiii

INTCAL98 Radiocarbon Age Calibration

Stuiver, M., et. al., 1998, Radiocarbon 40(3), p1041-1083

Mathematics

A Simplified Approach to Calibrating C14 Dates

Talma, A. S., Vogel, J. C., 1993, Radiocarbon 35(2), p317-322

Beta Analytic Radiocarbon Dating Laboratory

4985 S.W. 74th Court, Miami, Florida 33155 • Tel: (305)667-5167 • Fax: (305)663-0964 • E-Mail: beta@radiocarbon.com

APPENDIX B – HISTORIC DOCUMENTS (From State Archives)

<u>DOCUMENT</u>	<u>PAGE</u>
Minutes from a Privy Council meeting on May 31, 1851 (see Resolution 11)	39
Letter dated June 3, 1893 from M.H. Reuter to Minister of Interior J.A. King	41
Letter dated January 8, 1894 from August Unna to Minister of Interior E.O. Hall	42
Letter dated April 25, 1876 to D. Uaiwa from Minister of Interior W.L. Moehonua	43
Letter dated January 8, 1894 from Unna to Minister of Interior L.G. Wilder	44
Department of Interior document dated July 21, 1893 by G.L. Brown	45

Privy Council May 31 1861

Present His Highness John Young

Paki	Mr. Judd
Kekuanaka	, Wyllie
Kanaina	, Armstrong
Ii	Kanoa
Kaee	Kapeau
Kapaakea	

Lorrin Andrews Secretary

Prayers were read
Minutes were read and approved

Mr. Judd

Duties on Mission-
ary Importa-
tions

Mr. Judd read some communications from the
Collector of Customs in relation to payment of duties
on goods imported by the American Missionaries. -
after conversation Mr. Wyllie was appointed a Committee
to confer with Mr. Castle on the subject

Cooke & Castle
Titles to
Missionary
Lands

Mr. Armstrong read a letter from Messrs. Cook and
Castle relative to the title of Missionaries to their
dwelling houses and lots upon which the following
Resolutions recommending a committee of three to be
appointed to report on the subject; was adopted.

Messrs. Wyllie, Bates & Lee were the committee

Res. 1

Res. 1.

Resolved; That the letter of Messrs. Castle &
Cooke of April 9th be referred to a committee, consis-
ting of Messrs. Wyllie, Lee & Bates, to report upon
at a ~~future~~ subsequent meeting of this Council.

Hoa.

Mr. Armstrong read the application of Hoa for an
allodial title to his land

Res. 2.

Res. 2.

Resolved; That the Minister of the Interior be
and is hereby authorized to grant Hoa a fee simple
title to the land of land in Puna, Honolulu,

of \$2.00 per acre, provided always that this grant is not in any way to interfere with the land sold to Kaneihalaui and others.

✓ Konaiaiale

Mr. Armstrong read an application of Konaiaiale for an allodial title

Res. 9.

Res. 9.
Resolved; That the Minister of the Interior be and is hereby authorized to grant Konaiaiale a fee simple Title to the land awarded to him, and described in award No. 1552 of the Land Commission upon his paying the sum of Fifty Dollars for the Government rights therein.

↓ School House lot.

Mr. Armstrong read an application for a school lot in Kula o Kahua

Res. 10.

Res. 10.
Resolved; That this council approve of the selection by the Minister of Public Instruction of Lot No 6 Kula o Kahua and hereby authorize the Minister of the Interior to reserve said lot as a site for a school house, it being the only lot unsold between Alapai and Washington Streets.

Needham & Kelk

Mr. Armstrong read the application of Messrs. Needham & Kelk for land in Hana

Res. 11.

Res. 11.

Resolved; That the Minister of the Interior be and is hereby authorized to grant to W.G. Needham & E.J. Kelk a fee simple title to 441 acres of land in East Honoumaele Hana; Maui as surveyed by Mr. Alexander, for the sum of three Dollars per acre, reserving the rights of native tenants.

✓ Kupa & Makahiki

Mr. Armstrong read an application from Kupa and Makahiki for land in Hana

Honolulu
Hana Maui June 3-1893

His Excellency J. A. King
Minister of Interior
Honolulu H. I.

Sir:

I heard that Mr F. Wittrock has give up and
returned his lease of government land known
Kawela Horomau and Kaulaue at Hana Maui
to the Government if it so. I will offer you One
Hundred Dollars for year rent term of fifteen
year.

I remain

Your Obedient Servant
M. H. Reuter

1893, July
Reproduced on
State Archives copier

Description for lease of Govt land in Honomaele
and Kaeleku, Hana, Maui

All that Govt land in Honomaele, Kaeleku and Kaeleku,
Hana, Maui lying within the following boundaries:

Beginning at Govt Survey Station on Hill Olopana,
hence running by true bearings:

$62^{\circ} 00' W$ 6500 ft more or less to gulch boundary between
East and West Honomaele.

Hence along said gulch to Sea Shore.

From Initial point again:

$\sqrt{51^{\circ} 00' E}$ 3500 ft to boundary of Grant 766 (Chas Copp)

Hence along Grants 766, 2935, and 2640 to Port Road

Hence eastward along mauka side of Govt Road a
distance of 1020 ft.

Hence $N 30^{\circ} 15' E$ 5200 ft more or less to Sea shore,
and along shore to mouth of gulch, boundary between
east and west Honomaele.

Over 1250 Acres exclusive of
various Grants lying within these boundaries.

J. H. Brown
July 2nd 1893

Dollars

Hana January 8th 1874

His Excellency E. O. Hall
Minister of Interior
Honolulu

Sir -

The undersigned herewith
encloses an order on Thos. H. Davies
for \$ 50 = being for Rent of Government
land at Kaeleku, Kawela & Honomaale
for two years ending August 1873; in
reference hereto please see your predecessors letters
dated Aug 21 - 1871 & Aug 26th 1872 -

I did not release the land above mentioned
because in 1873 I became owner of 600
acres of land at Kaeleku -- If Your
Excellency should be pleased, to lease
the said land belonging to Government at
Kaeleku, Kawela & Honomaale, situated
between the Government road & the sea
for the term of ten years, at a yearly
rental of Twenty Five Dollars as
charged by Your Excellency's predecessor, I
could wish to obtain a lease thereof.

Your Excellency's obedient servant Alano

April 25th 1876

April 25th 1876.

Uaiwa
D. Uaiwa

of Hana,
Paauhau, Maui.

Sir:-

Your letter requesting for a Royal Patent on the land of Honomale, awarded to Kahanu (k) under Award No. 42, has been received. Said Ahupuaa you say contain an area of 706 acres.

No Patent will be sent, if the Certificate of Award by the Land Commissioners is not produced at this office, the proceedings of which are recorded in the Government Records; together with the description of said land, and the record showing that the boundaries between Kahanu's portion and the Government's portion have been approved. If you will bring the same here, together with the appraisement made by the Appraisers to ascertain the Government's interest in all lands from Hawaii to Kauai, as prescribed by Law, the Patent will be promptly granted.

Yours respectfully

W. L. Moehonua

Minister of the Interior.

Hana January 9th 1880

S. G. Wilder Esq.
Minister of Interior
Honolulu

Indexed

Sir -

The undersigned, holding
Lease No 214 of Government lands at
Honomaale &c, paid Rent last
June 28 - 1878 for 2 years ending
December 31st 1879 & herewith takes the
liberty to enclose an order on
Messrs H. Hackfeld & Co, for \$25 -
covering Rent for the year 1880. -

Very Respectfully
Your Excellency's
obdt Servant

J. Uuna